

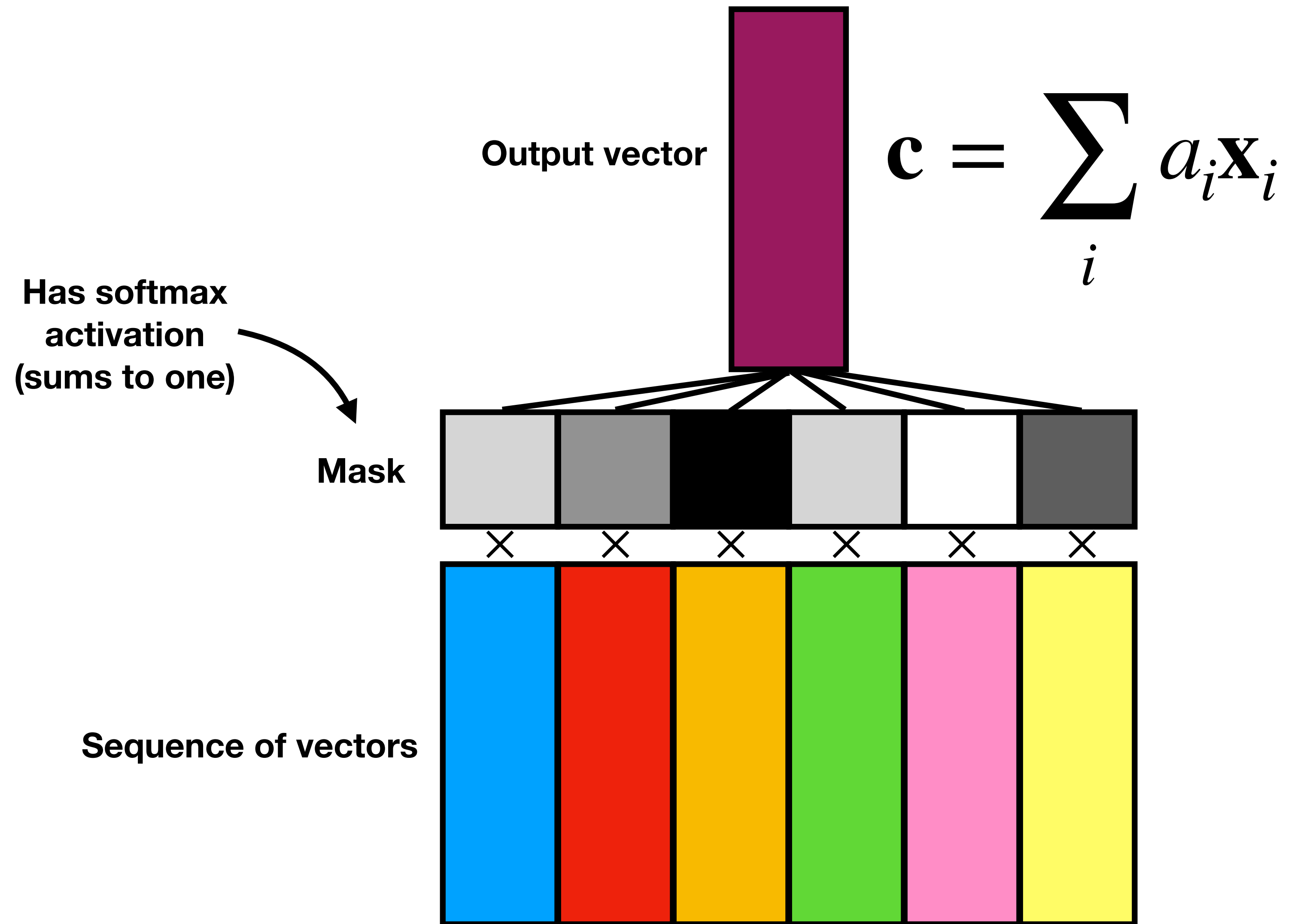
Attention

CSC 487: Deep Learning
Instructor: Jonathan Ventura

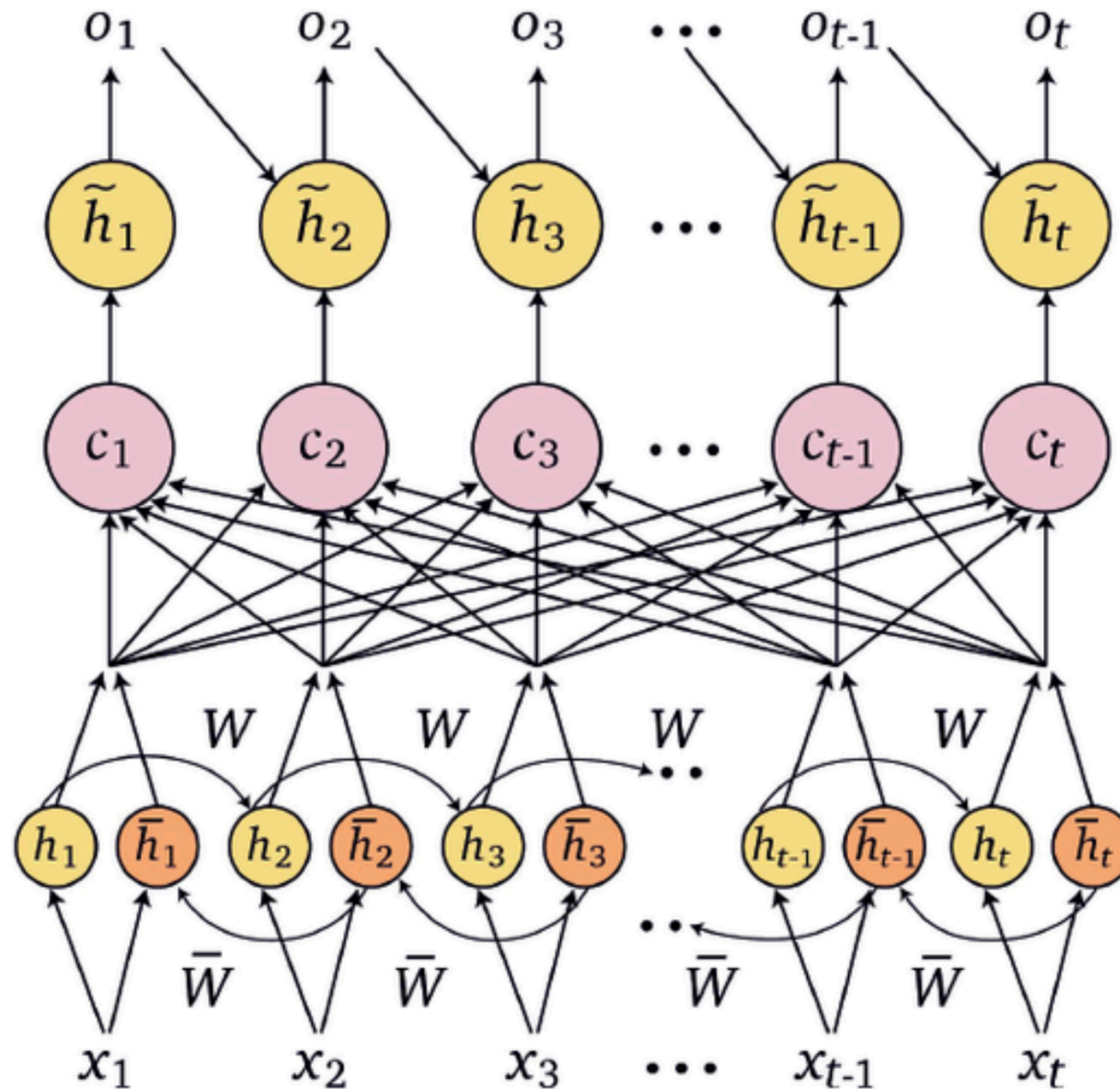
Outline

- What is attention?
- Use in sequence-to-sequence (encoder-decoder) model
- Scaled dot product attention
- Transformer encoder and decoder

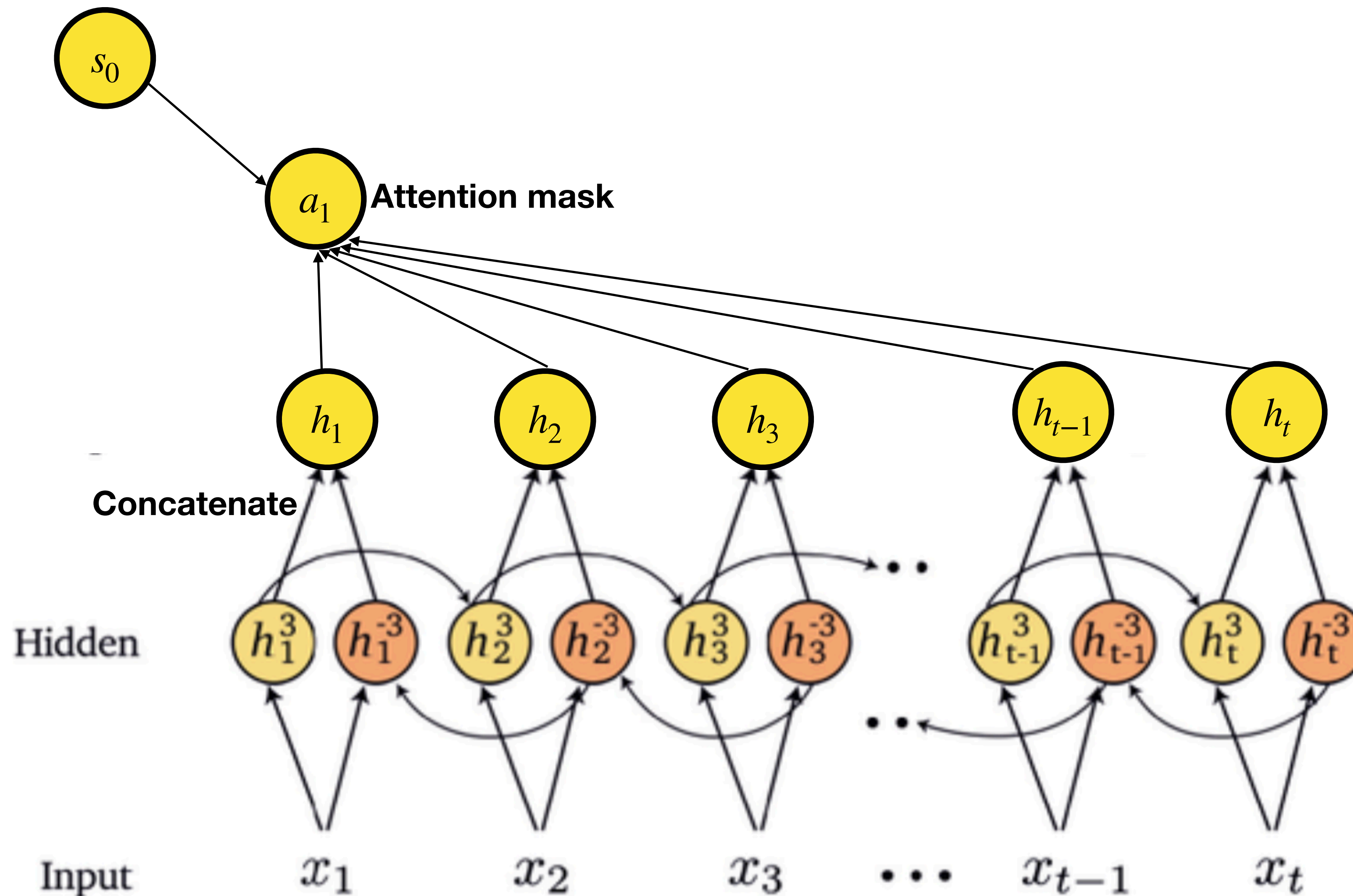
What is “Attention” ?

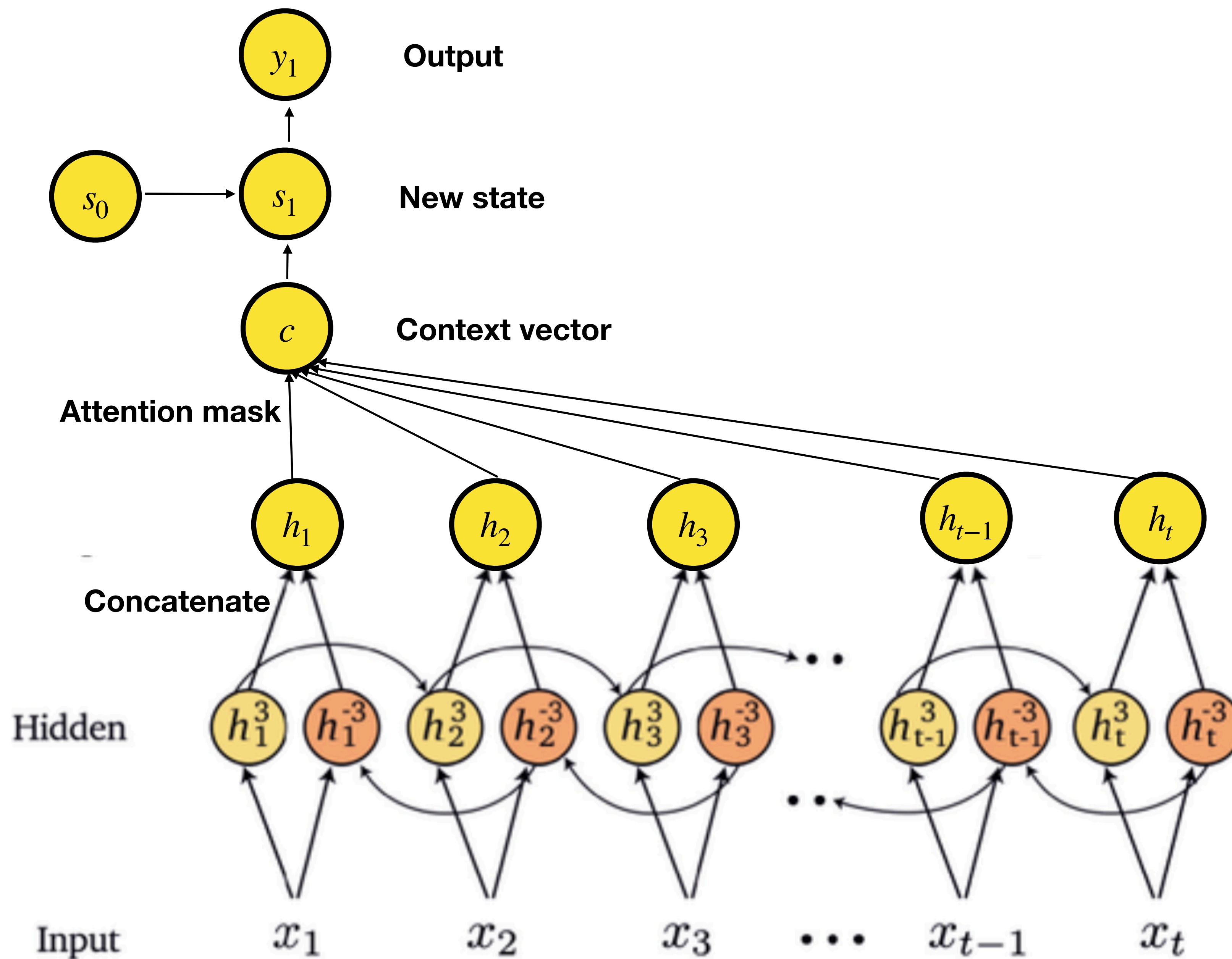


Sequence-to-Sequence with Attention

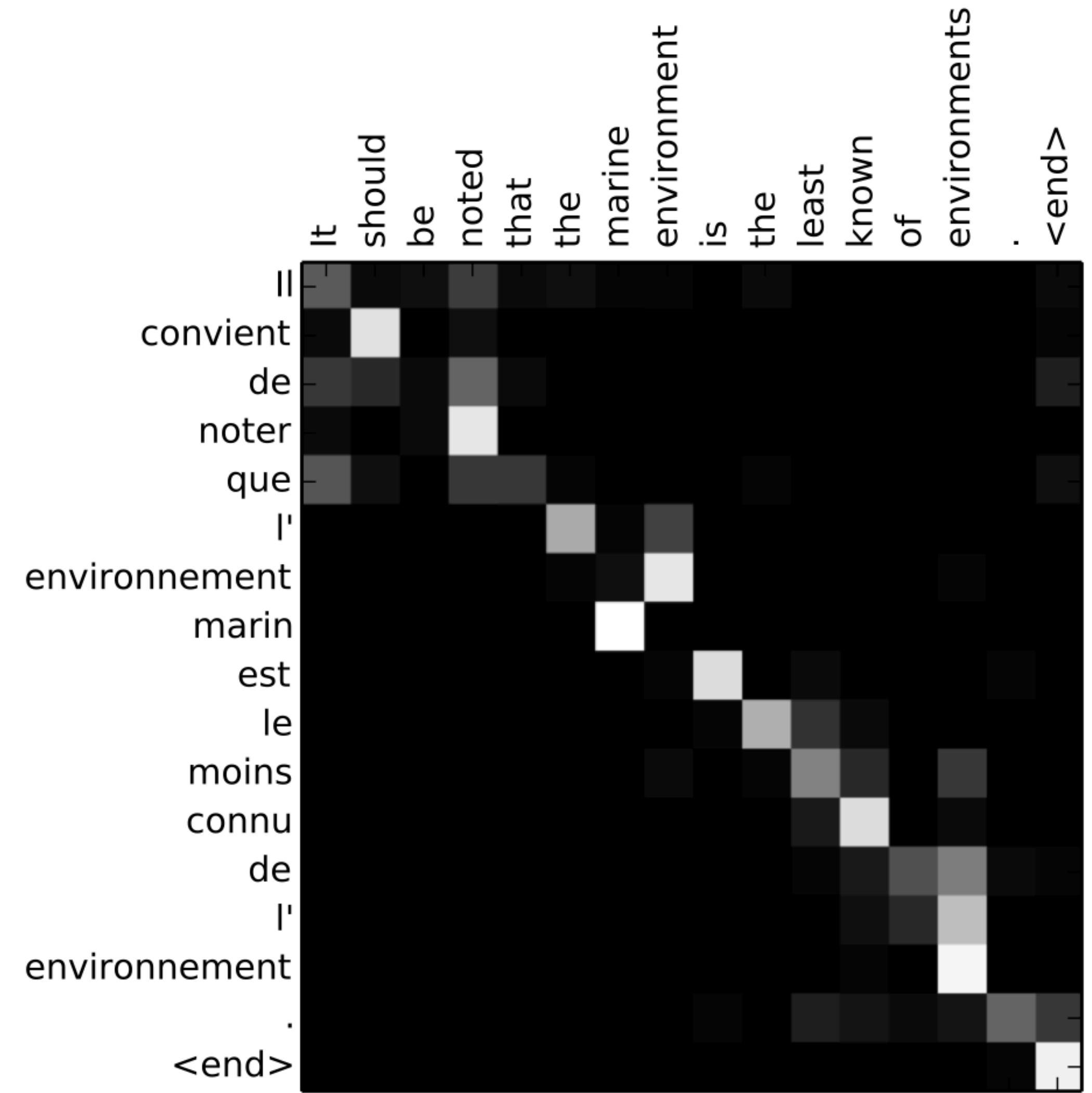
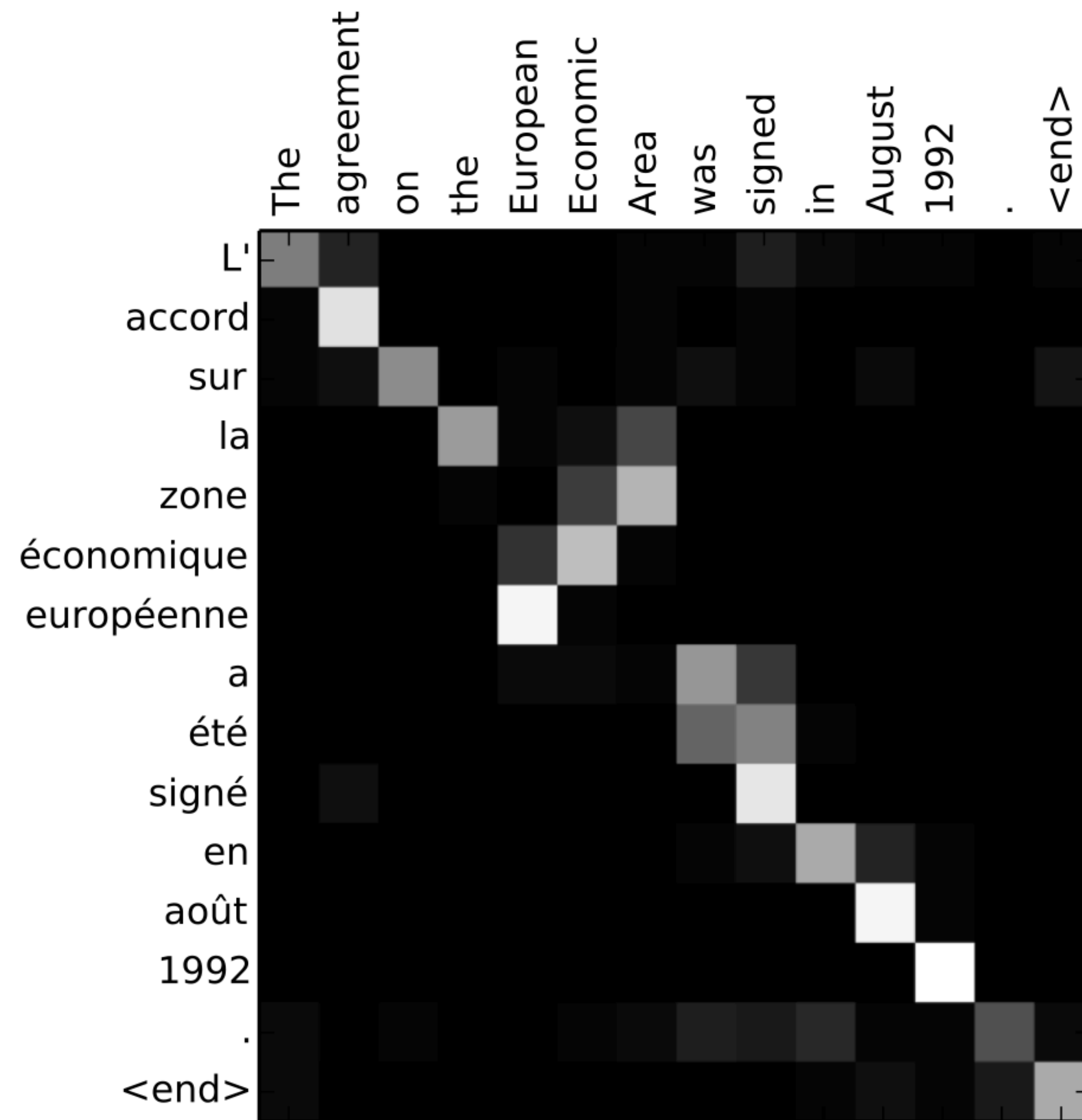


Attention Calculation





Translation Examples



Classification Examples

Negative Example

- if I can give this restaurant a 0 I will we be just ask our waitress leave because someone with a reservation be wait for our table my father and father-in-law be still finish up their coffee and we have not yet finish our dessert I have never be so humiliated do not go to this restaurant their food be mediocre at best if you want excellent Italian in a small intimate restaurant go to dish on the South Side I will not be go back

Positive Example

- love this place it really be my favorite restaurant in Charlotte they use charcoal for their grill and you can taste it steak with chimichurri be always perfect Fried yucca cilantro rice pork sandwich and the good tres lech I have had. The desert be all incredible if

Attention as “search”

Input text

The
dog
barks
and
chases
the
cat
.

← 90%

Output text

?

Attention as “search”

Input text

The
dog
barks
and
chases
the
cat
.

← 10%

Output text

?

Attention as “search”

Input text

The
dog
barks
and
chases
the
cat
.

← 0%

⋮

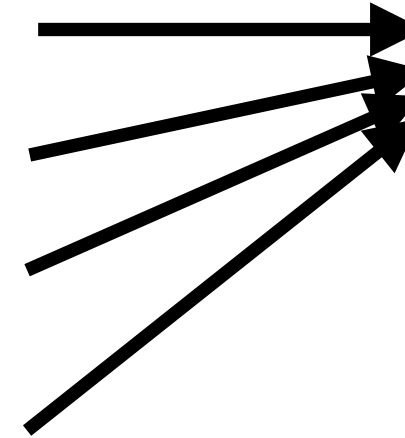
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⋮

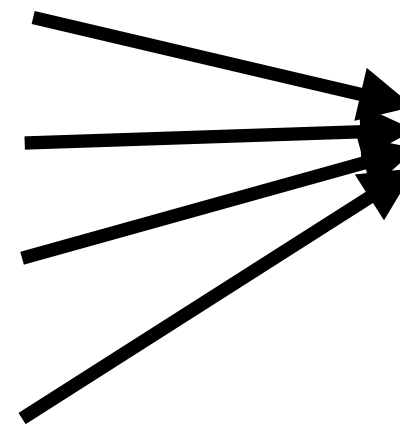
Output text

Le
?

Attention as “search”

Input text

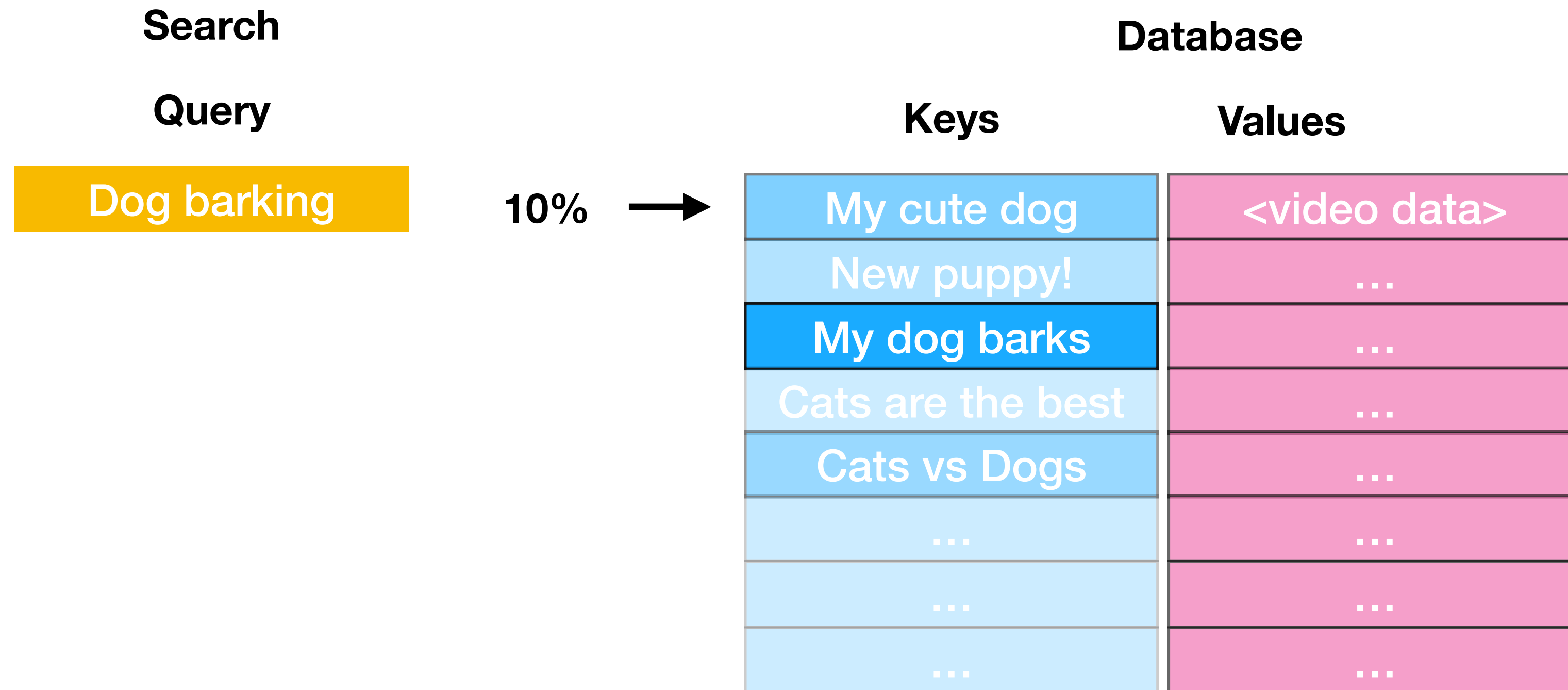
The
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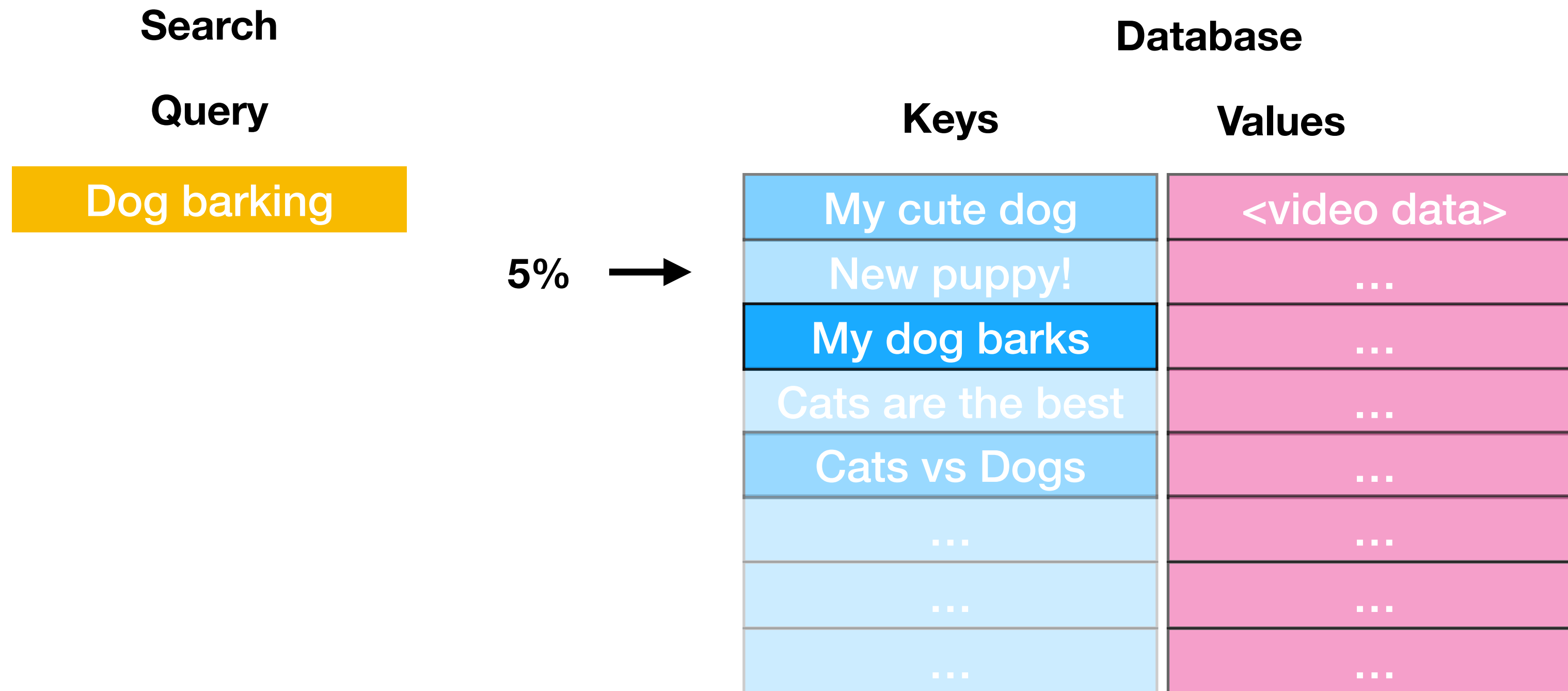
Output text

Le
chien

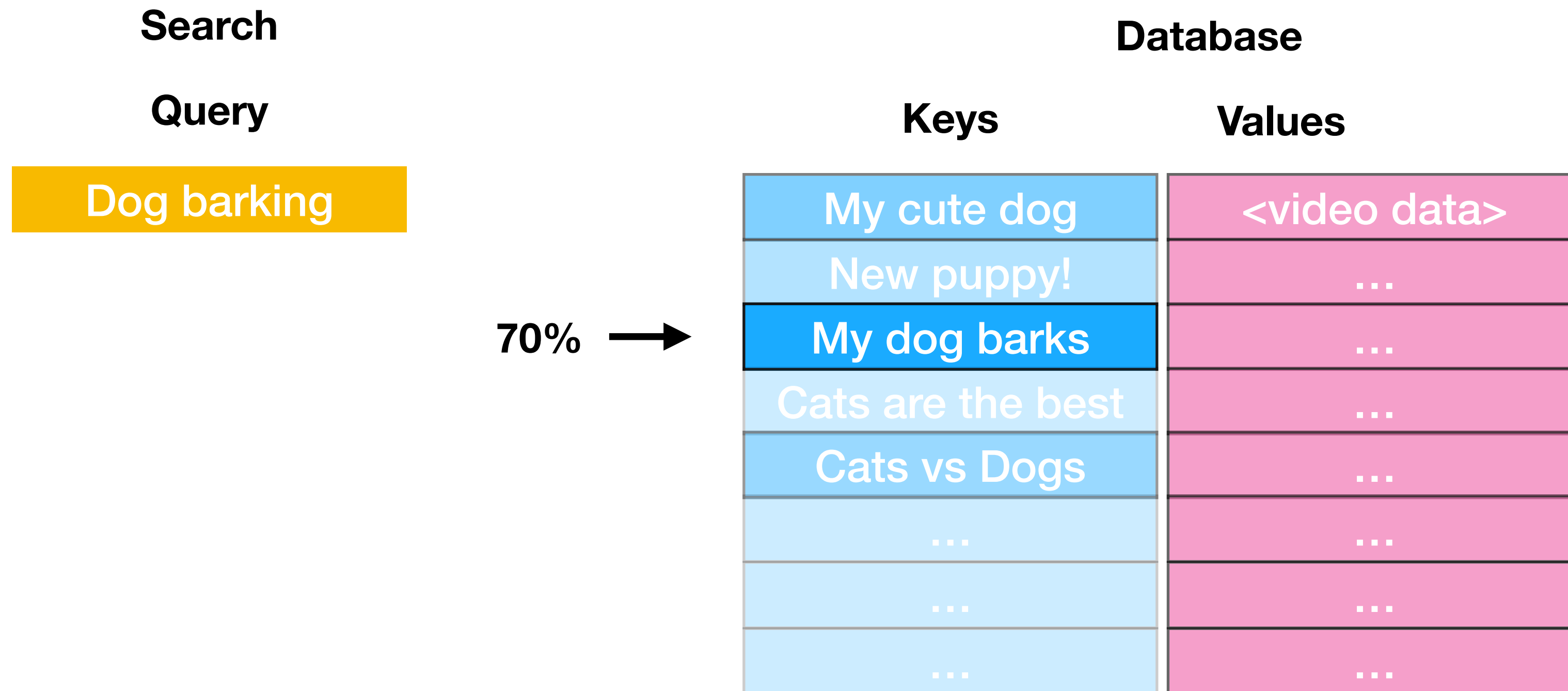
Query-Key-Value Attention



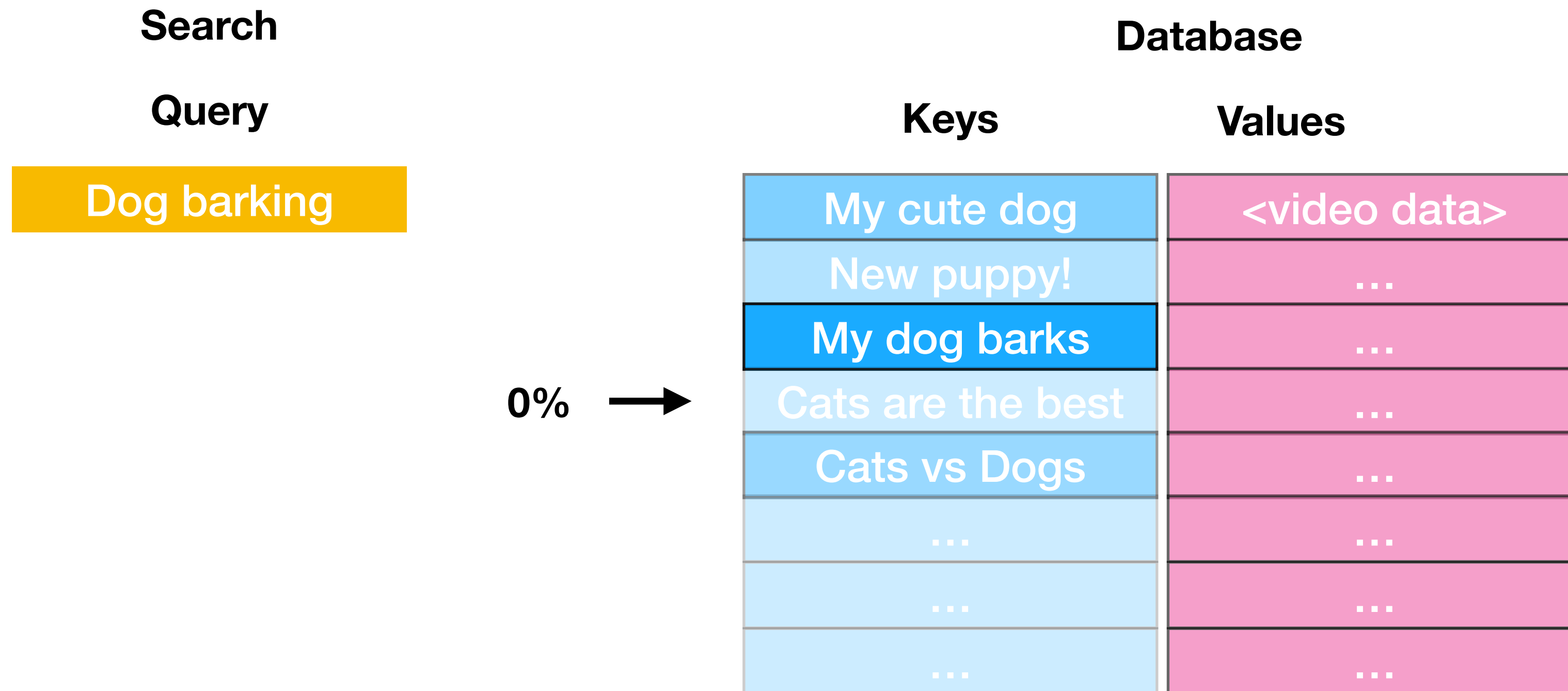
Query-Key-Value Attention



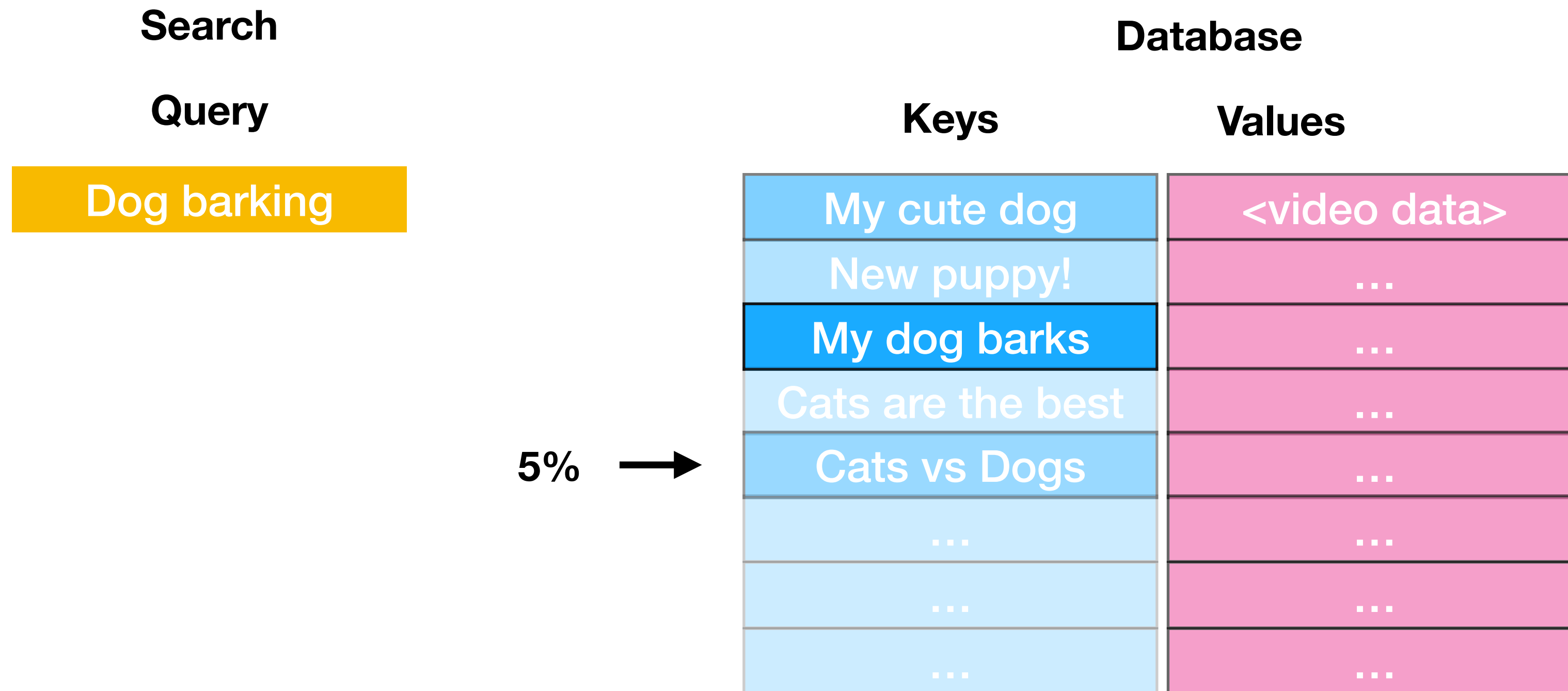
Query-Key-Value Attention



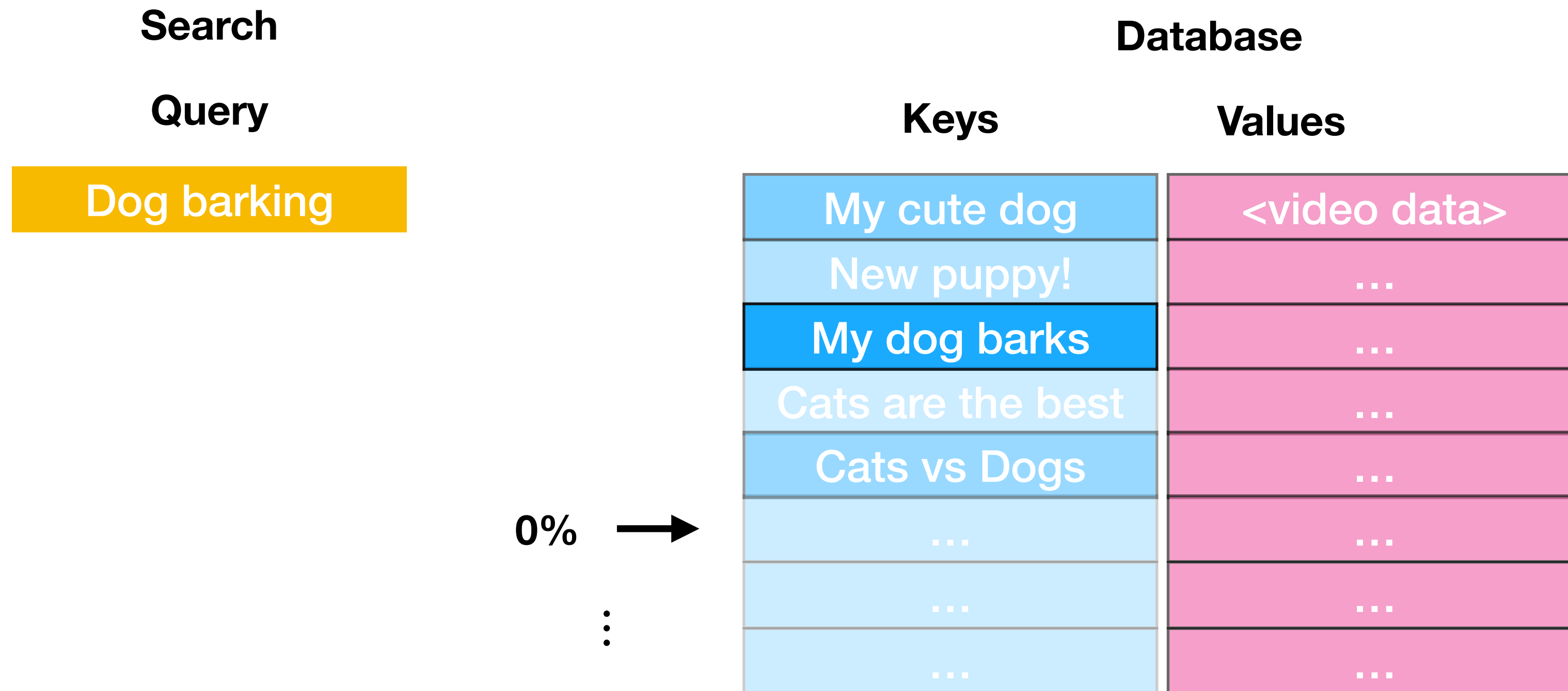
Query-Key-Value Attention



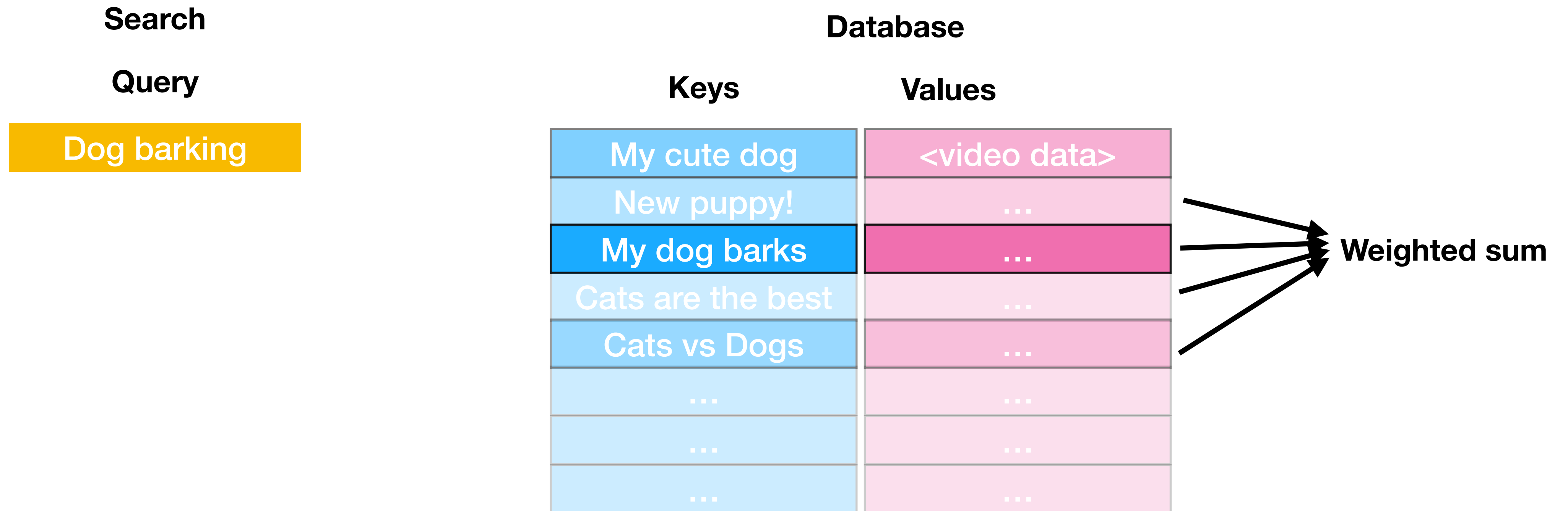
Query-Key-Value Attention



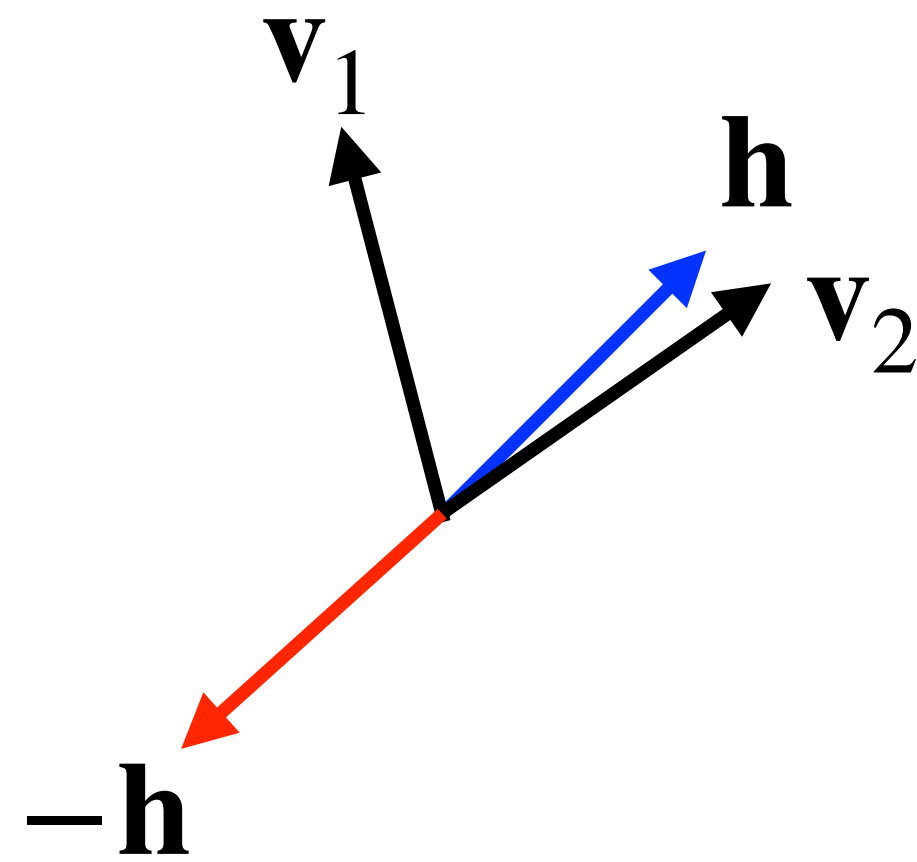
Query-Key-Value Attention



Query-Key-Value Attention



Vector Search by Dot Product



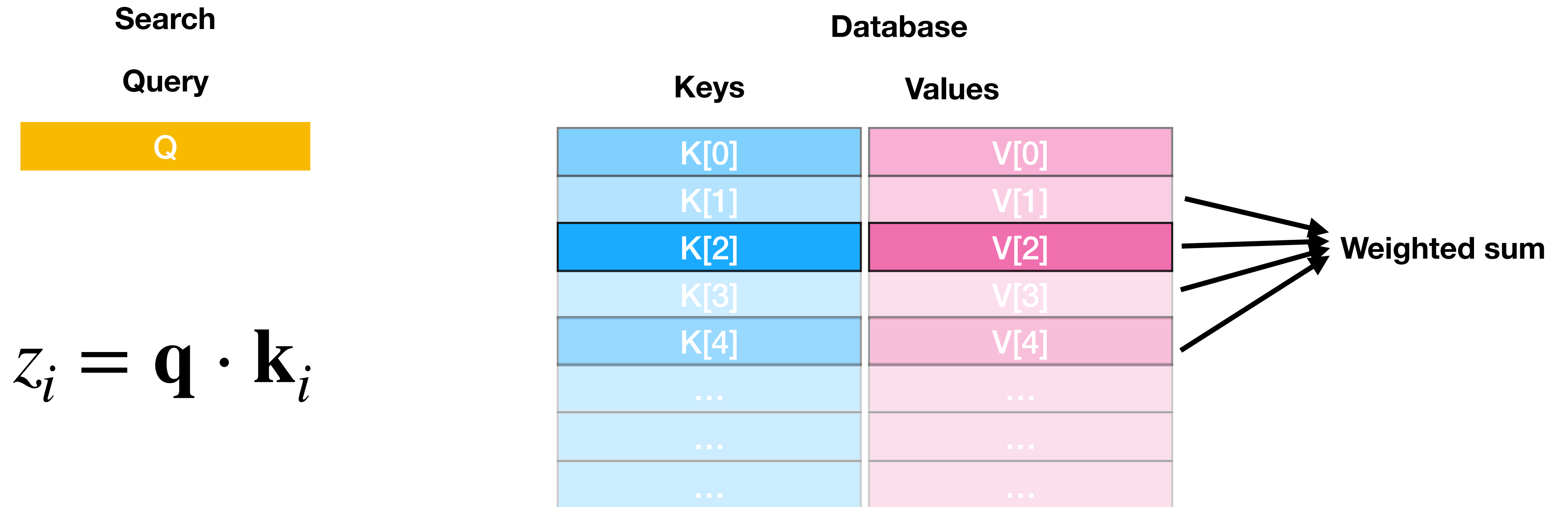
$$\mathbf{h} \cdot \mathbf{v}_2 > \mathbf{h} \cdot \mathbf{v}_1$$

$$\mathbf{h} \cdot \mathbf{h} = 1$$

$$\mathbf{h} \cdot -\mathbf{h} = -1$$

(if vectors are normalized)

Vector Search by Dot Product



Vector Search by Dot Product

Search

Query



Database

Keys

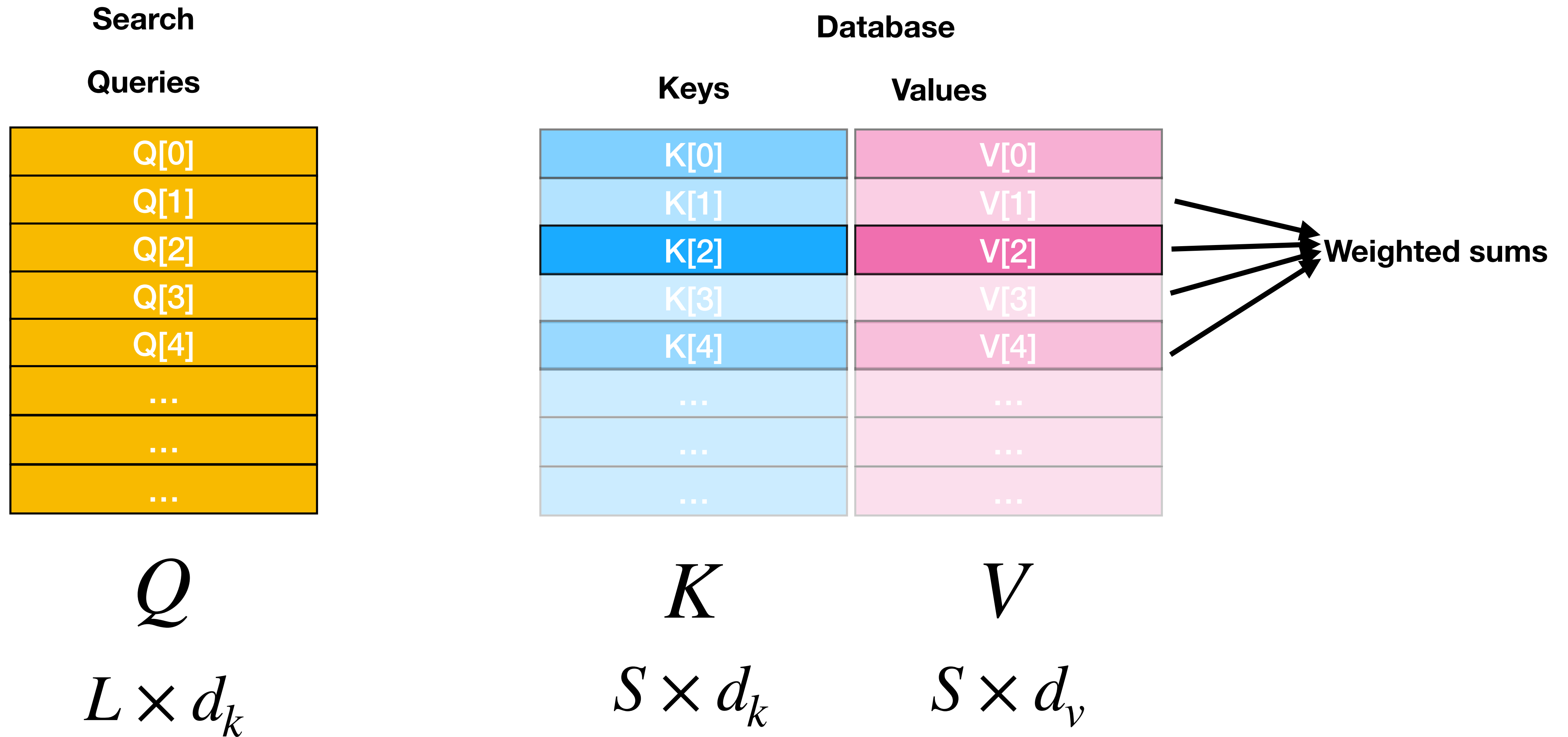
Values

K[0]	V[0]
K[1]	V[1]
K[2]	V[2]
K[3]	V[3]
K[4]	V[4]
...	...
...	...
...	...

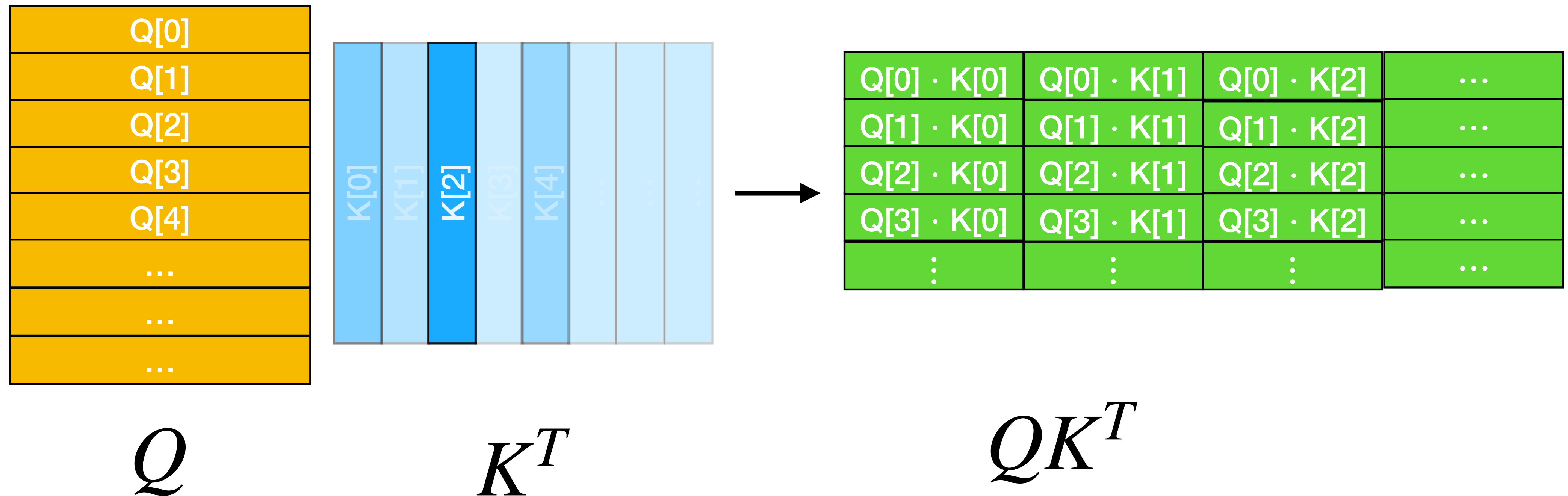


$$a_i = \frac{\exp(\mathbf{q} \cdot \mathbf{k}_i)}{\sum_j \exp(\mathbf{q} \cdot \mathbf{k}_j)}$$

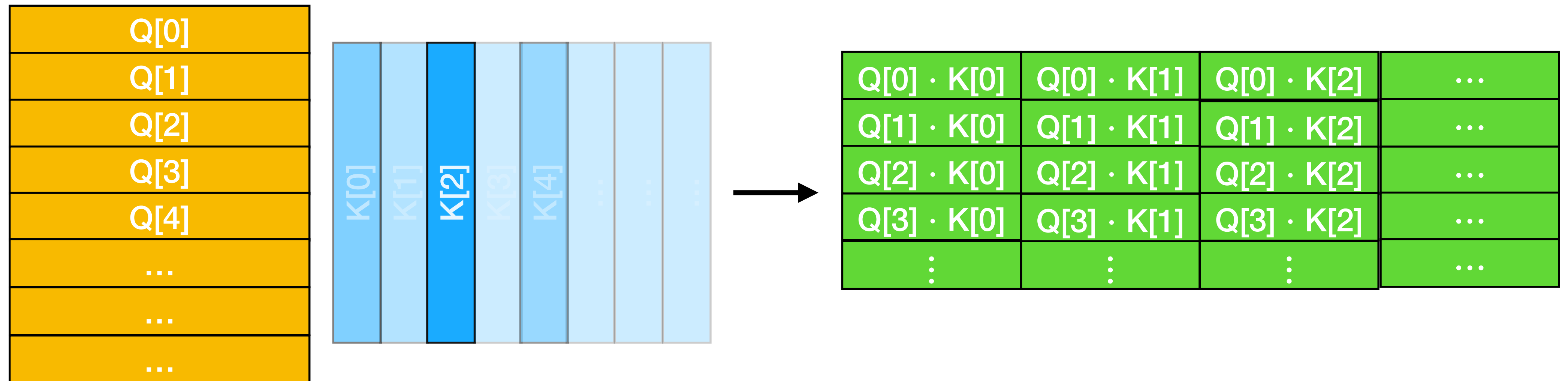
Multiple Queries



Multiple Queries



Multiple Queries



$$Q$$

$$L \times d_k$$

$$K^T$$

$$d_k \times S$$

$$QK^T$$

$$L \times S$$

Scaled Dot-Product Attention

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Q is matrix of “queries”

K is matrix of “keys”

V is matrix of “values”

d_k is dimension of queries and keys

Scaled Dot-Product Attention

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{\boxed{Q} \boxed{K^T}}{\sqrt{d_k}} \right) \boxed{V}$$

$L \times d_k$ $d_k \times S$ $S \times d_v$

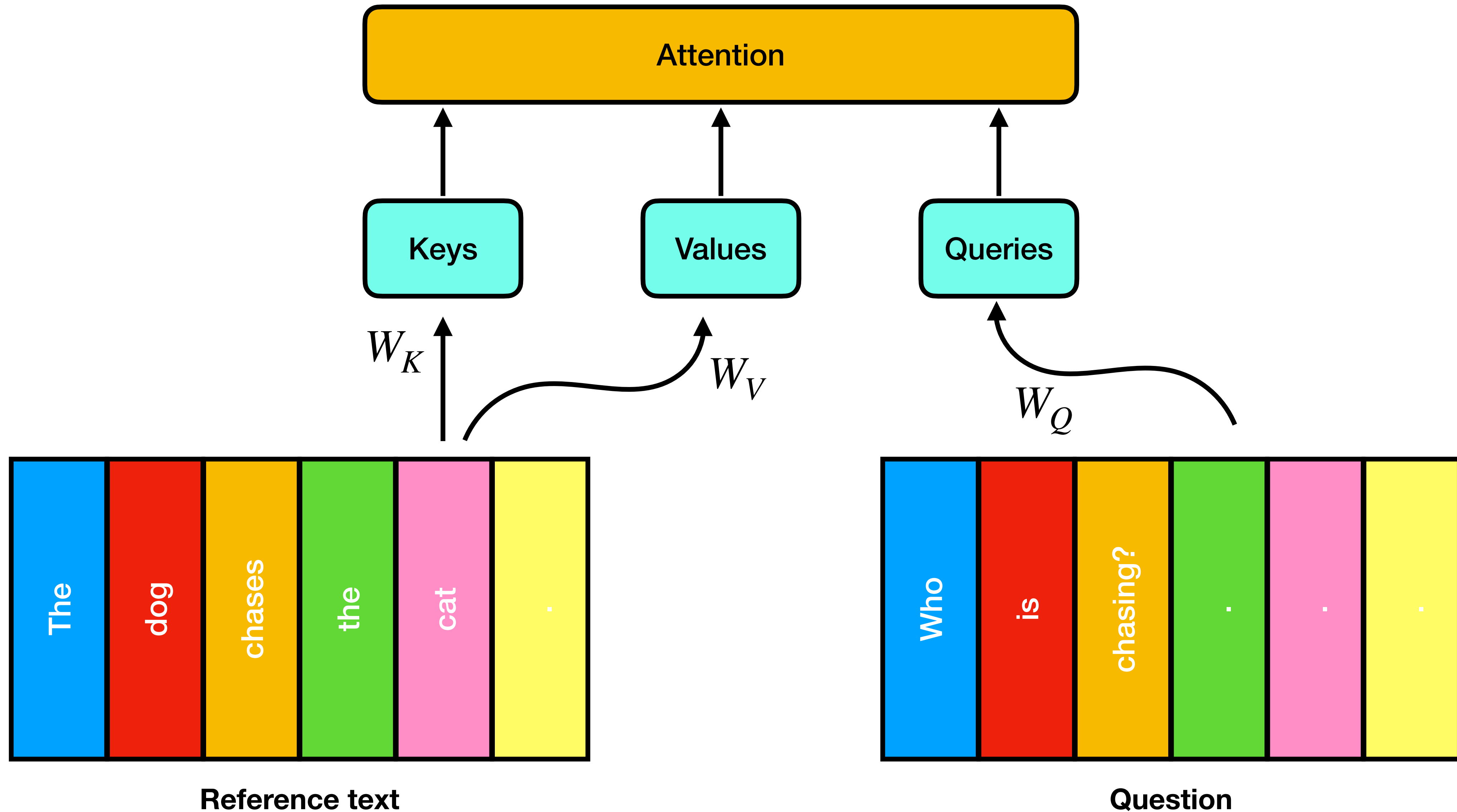
Sums to 1 along rows

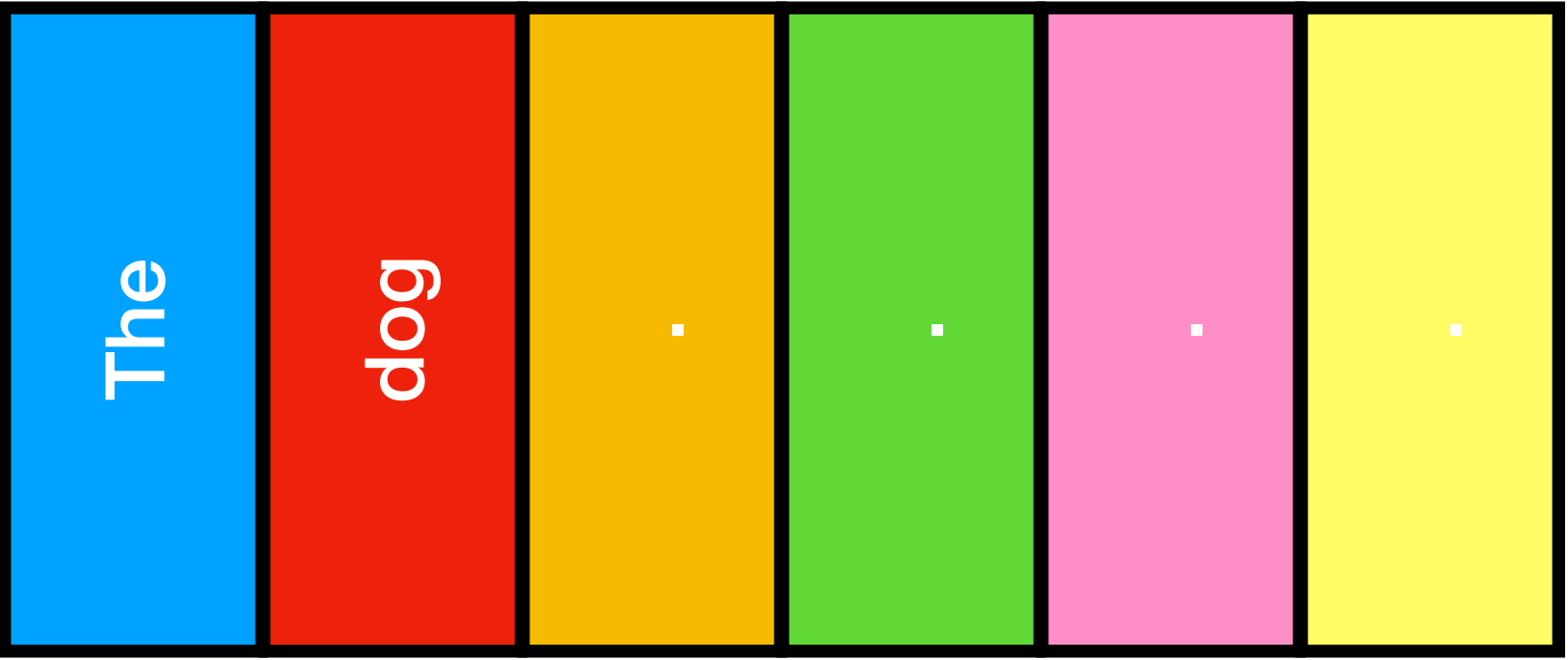
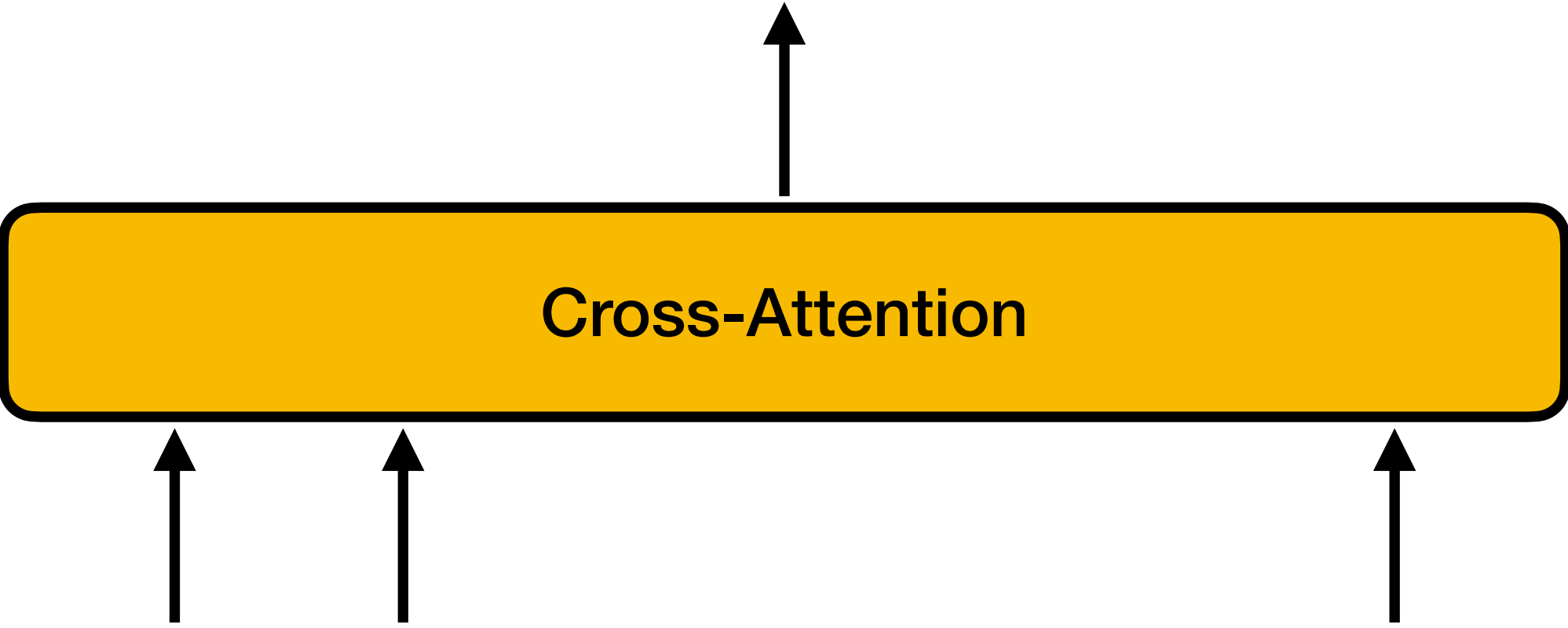
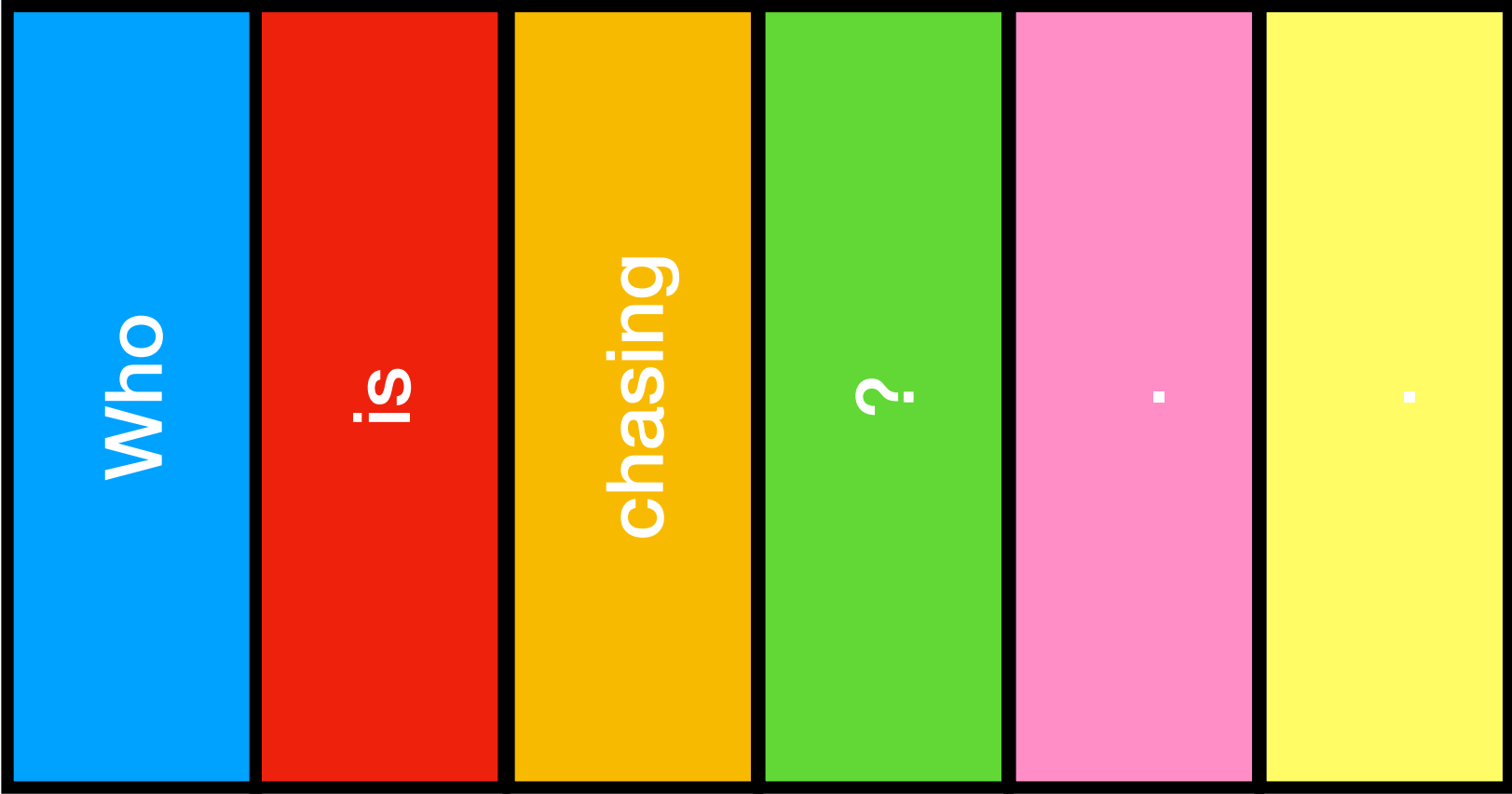
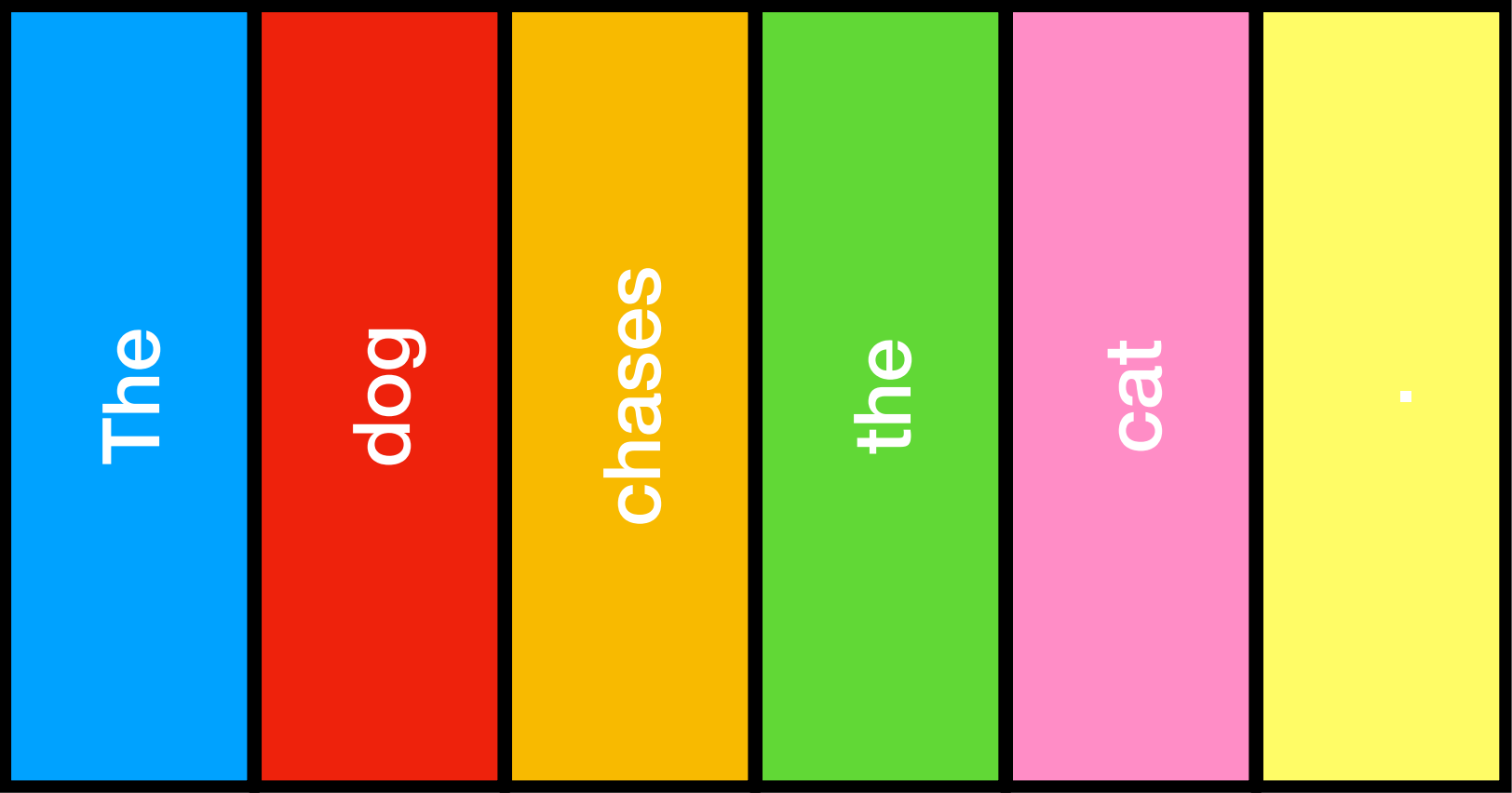
Retrieval System Analogy

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

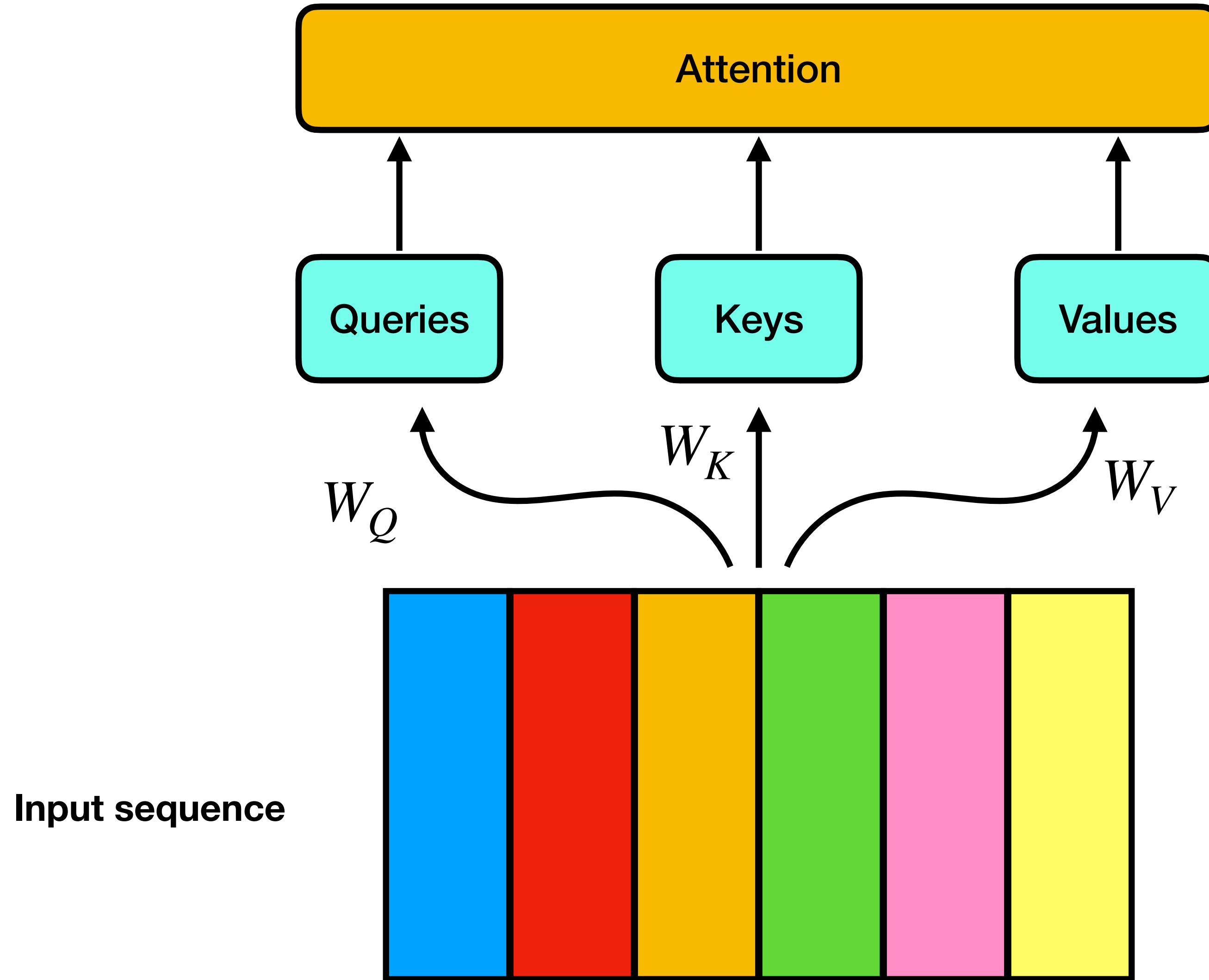
- Queries come from user (search terms)
- Keys are descriptors of data (database item IDs)
- Values are the data themselves (database items)

Cross-attention





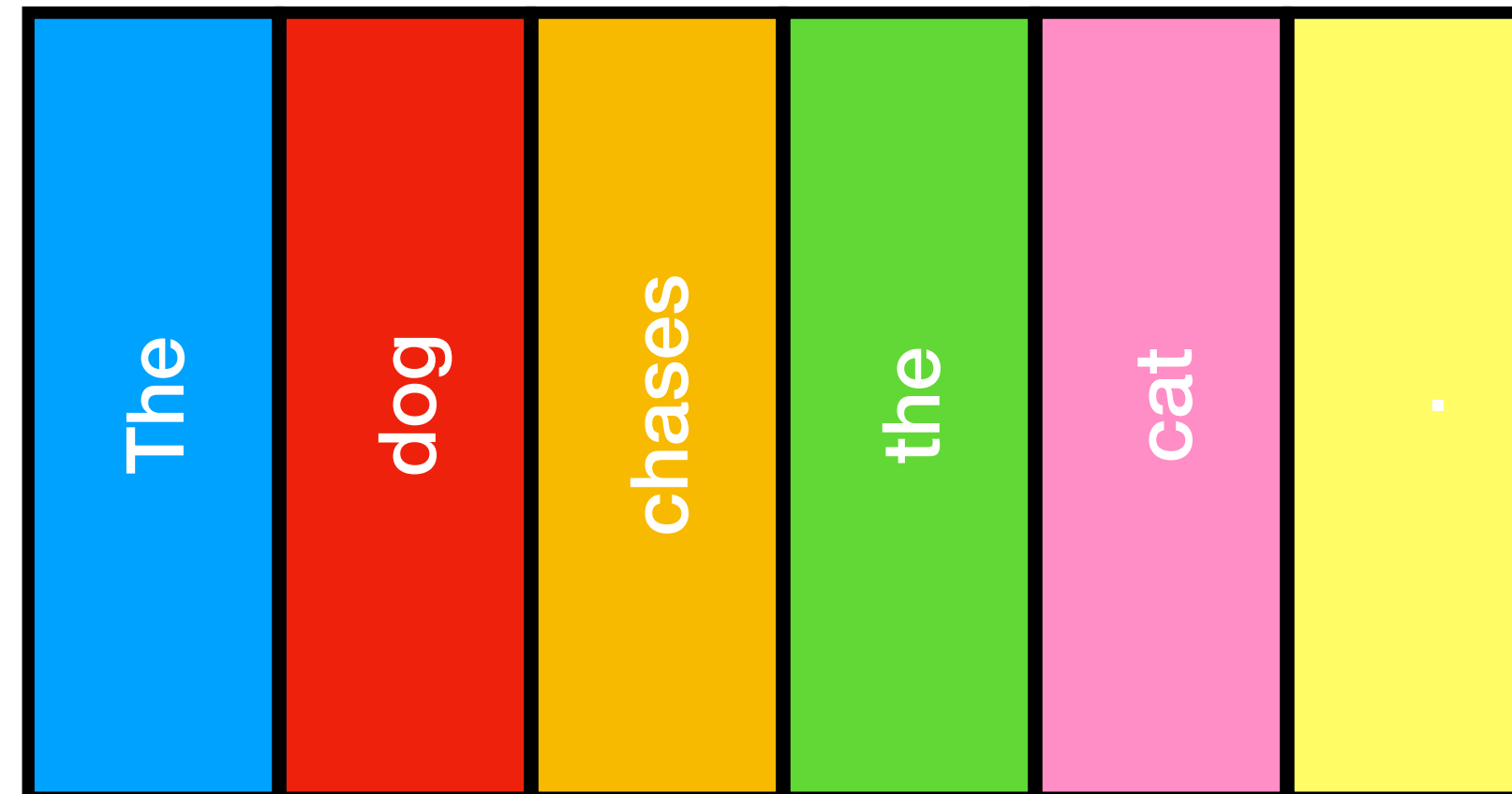
Self-attention



Output sequence

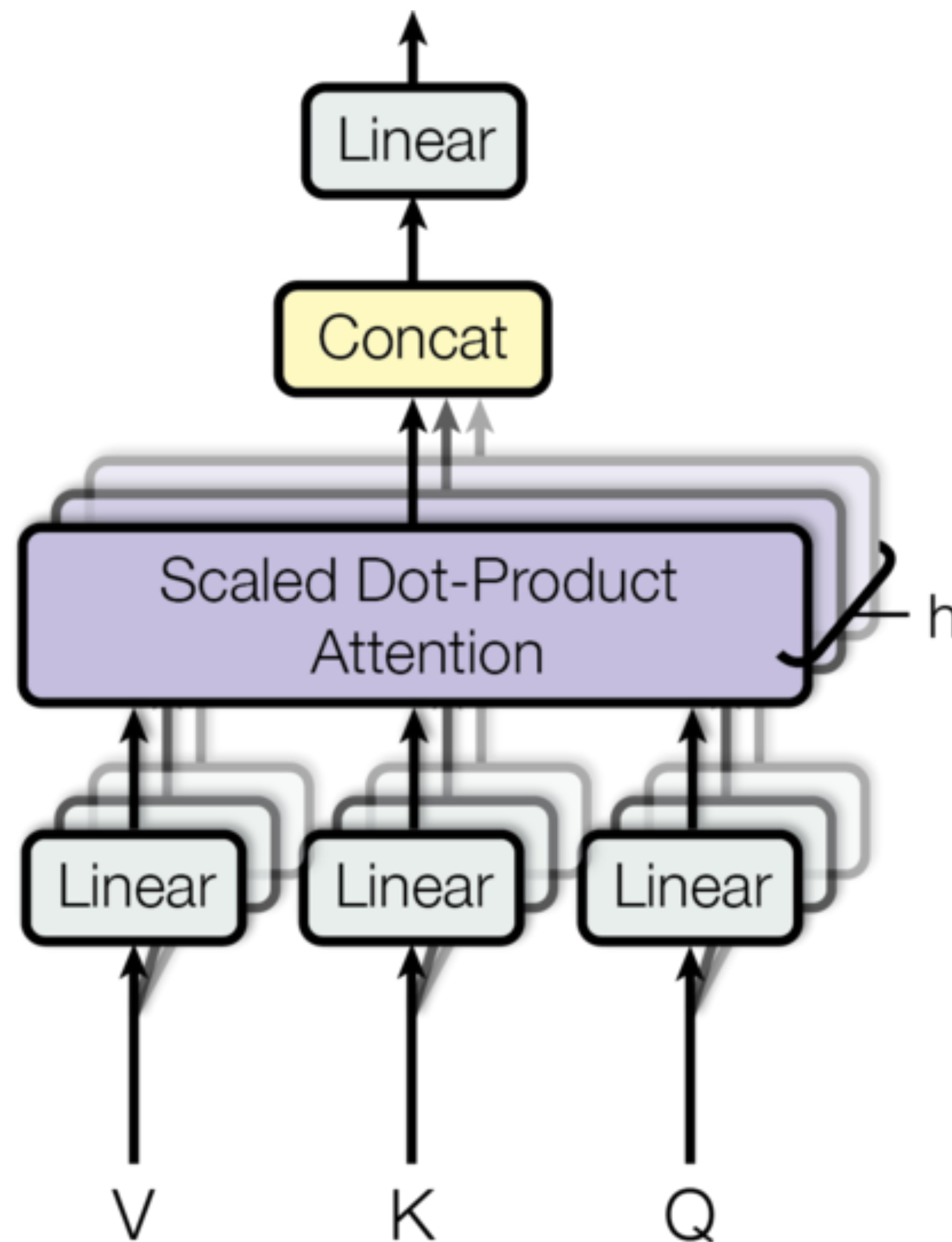


Input sequence

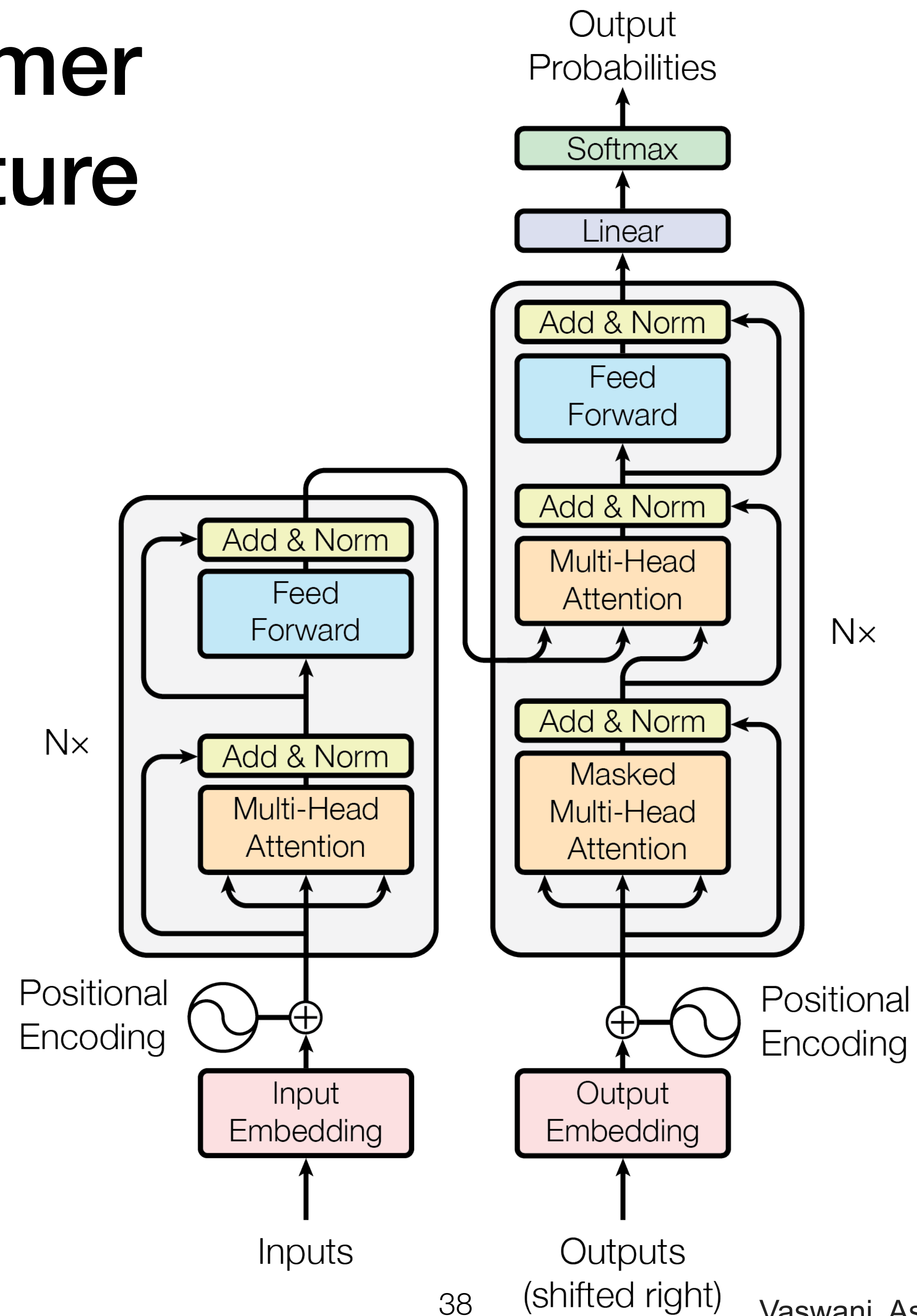


Multi-head Attention

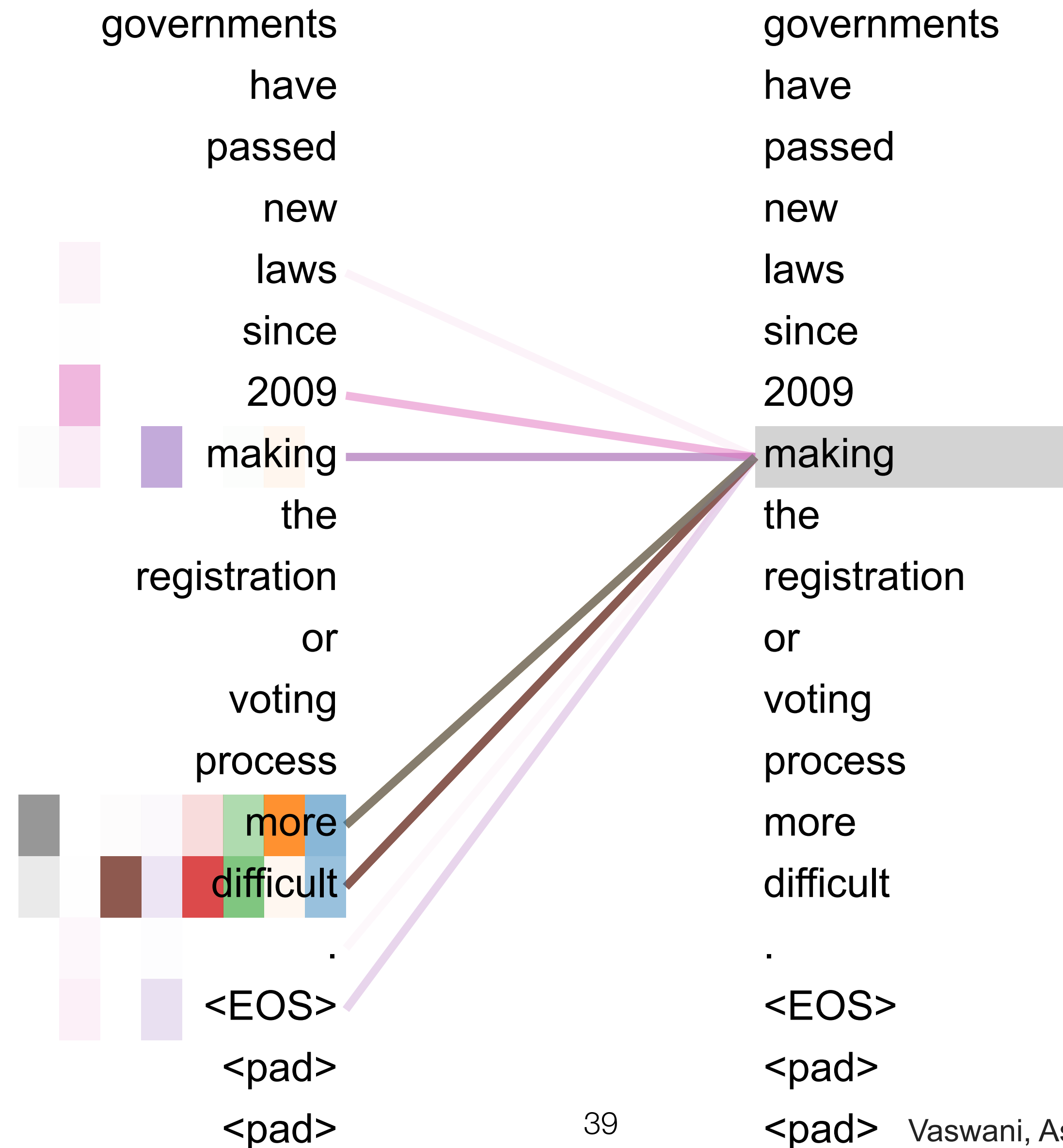
Each head has different projections of the Q,K,V

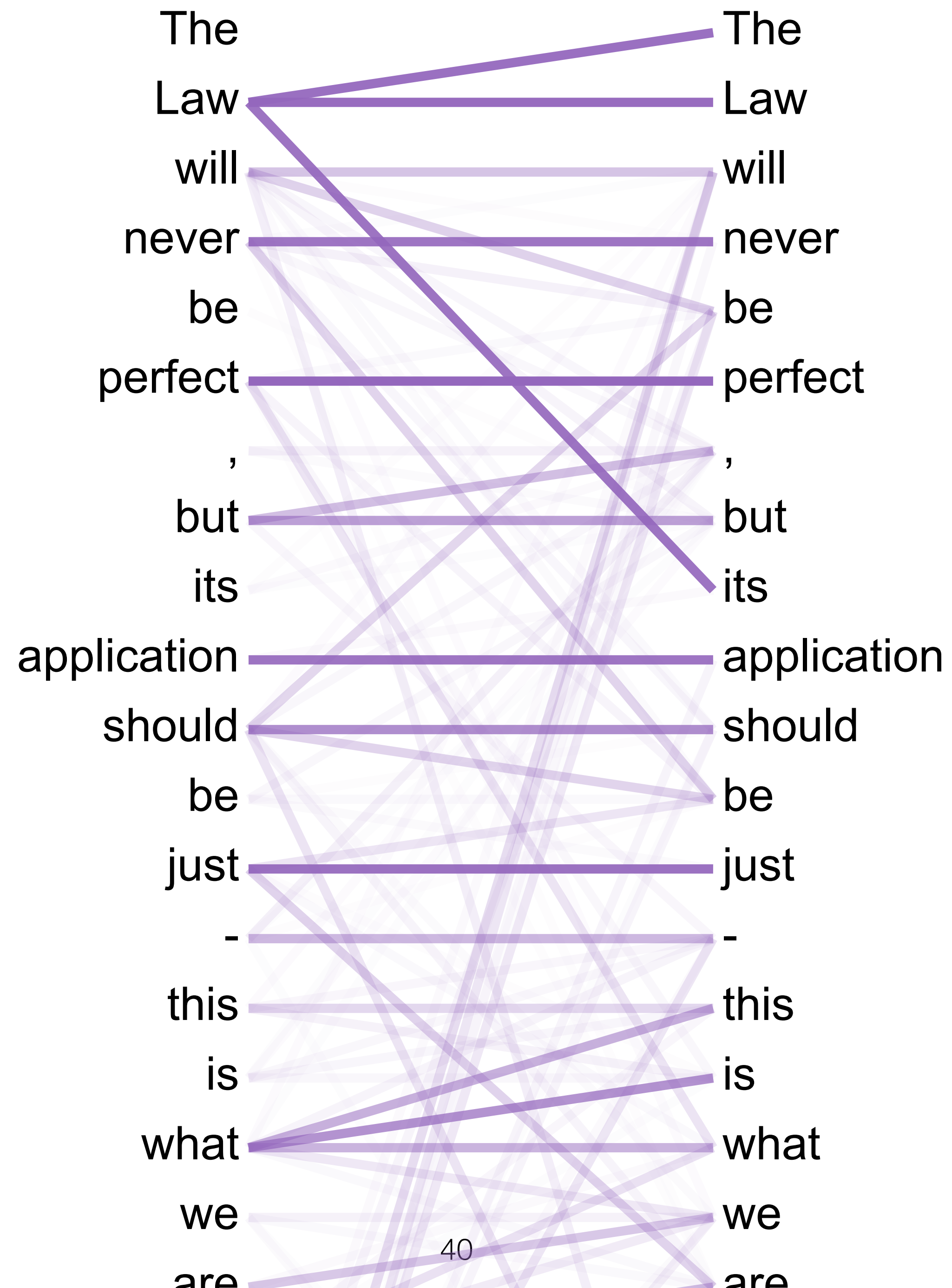


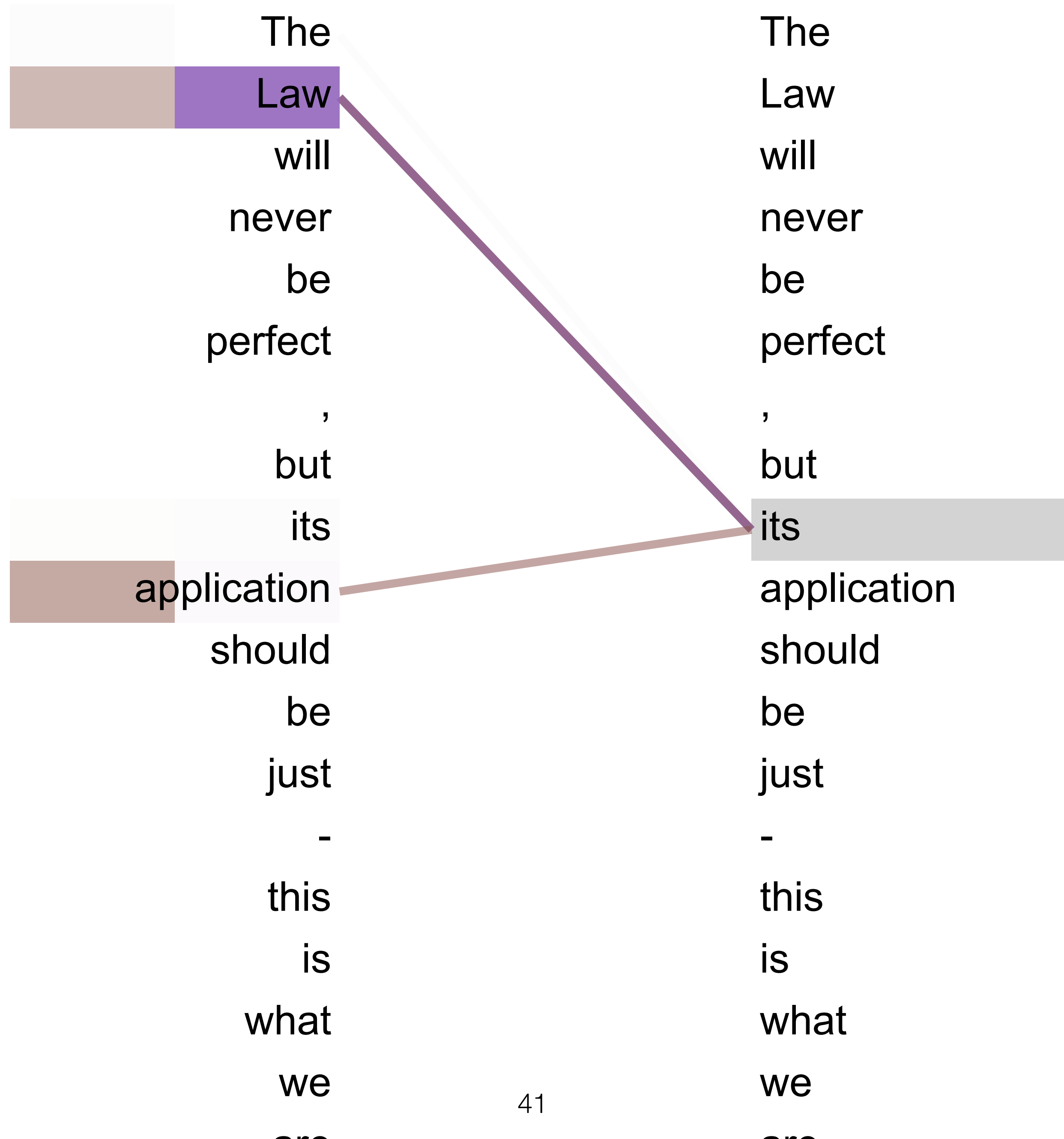
Transformer architecture



Self-Attention Visualization



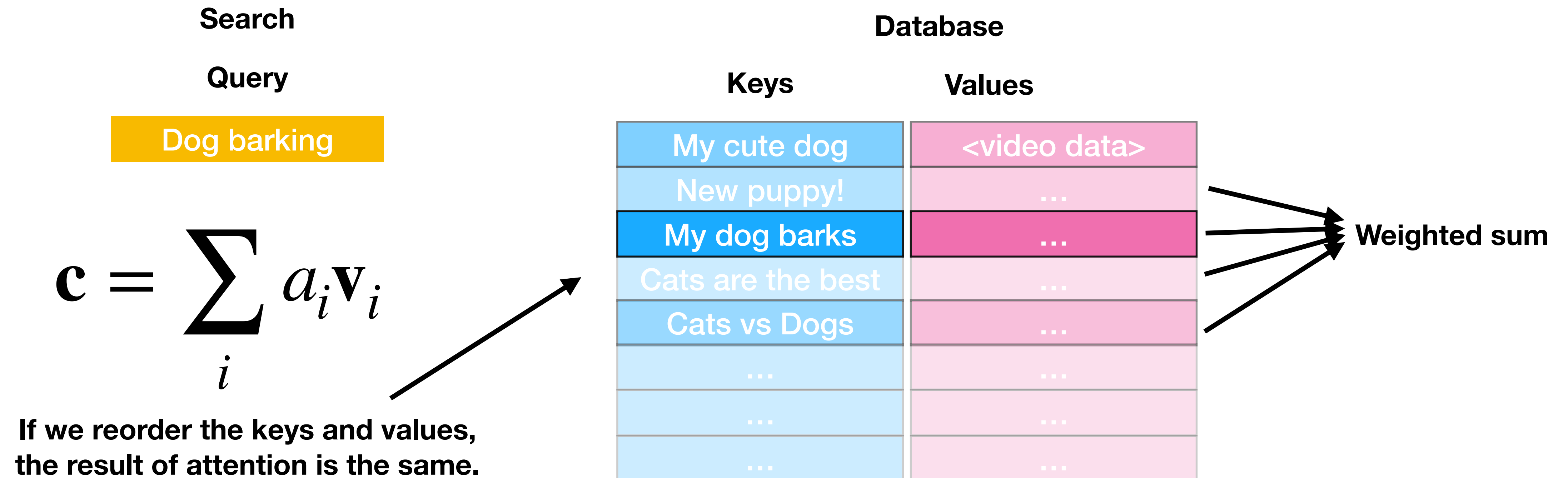




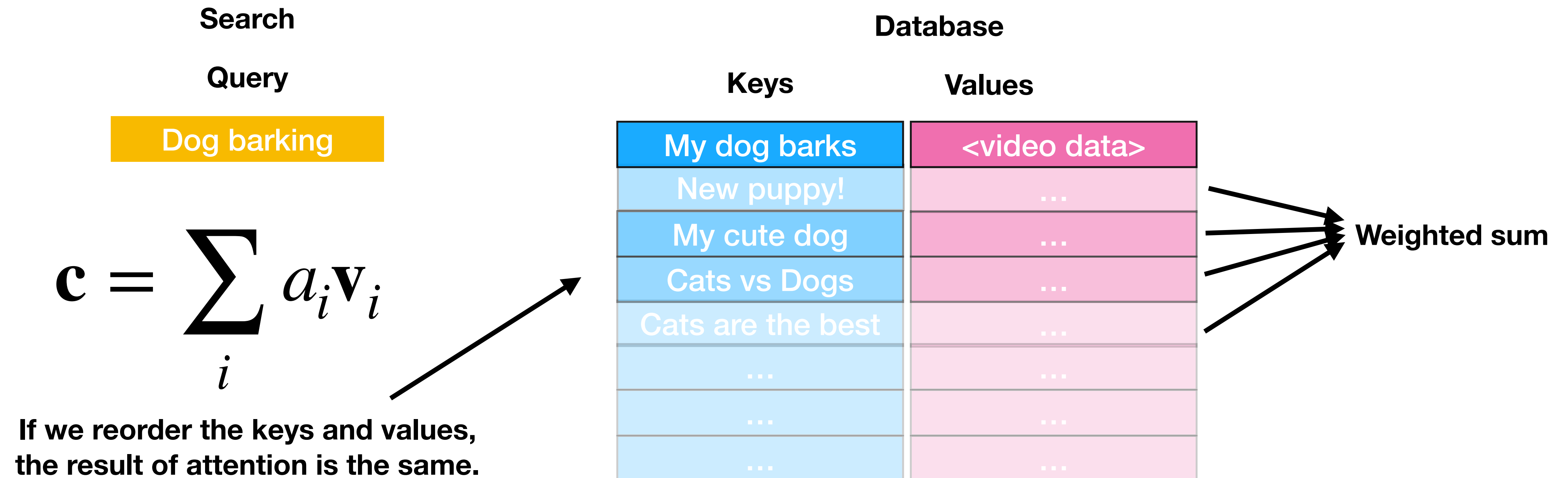
Permutation Invariance

- Attention block is “permutation invariant” meaning it doesn’t capture the order of the tokens in the sequence. Why?

Permutation Invariance



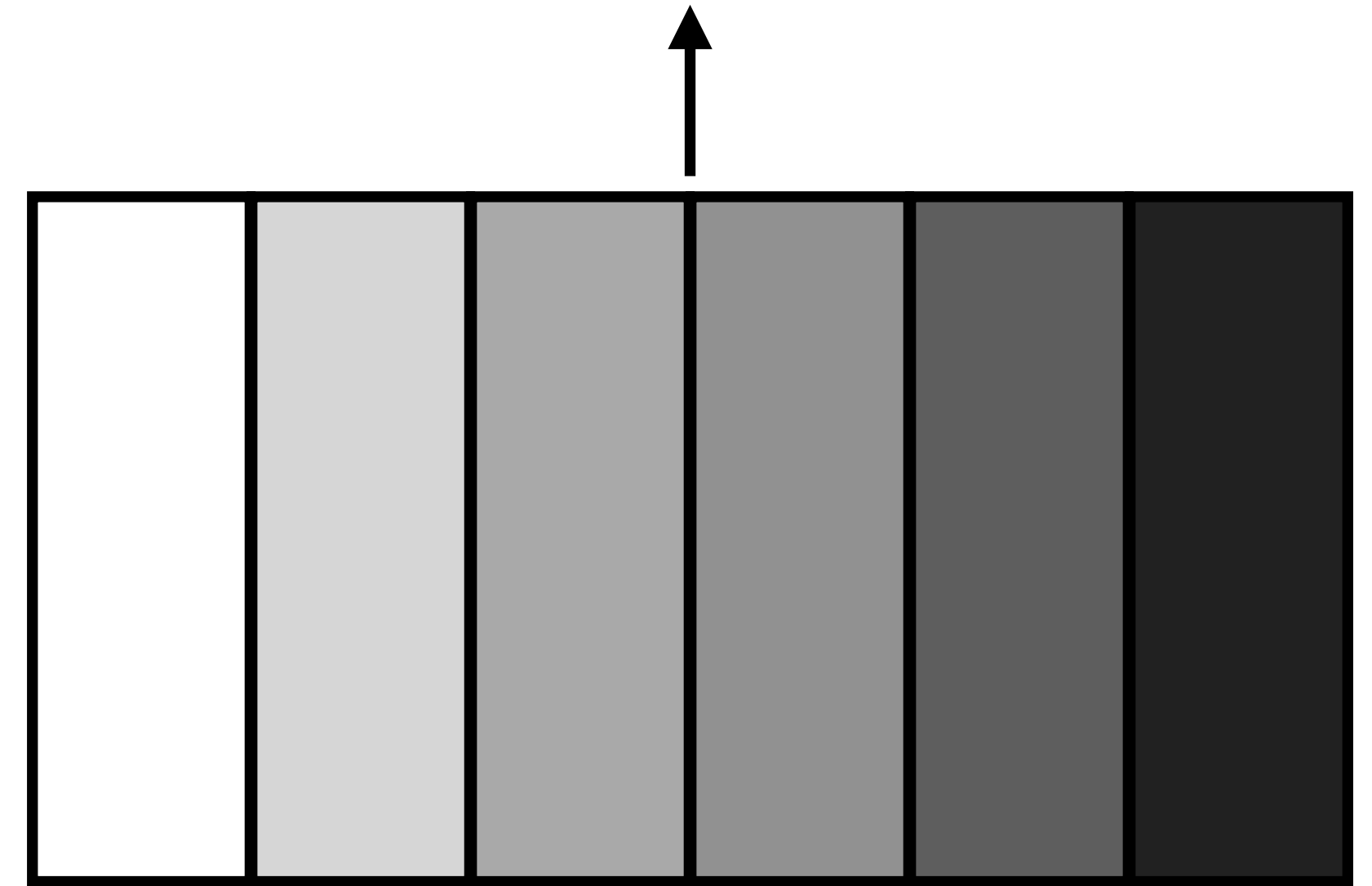
Permutation Invariance



Positional Encoding

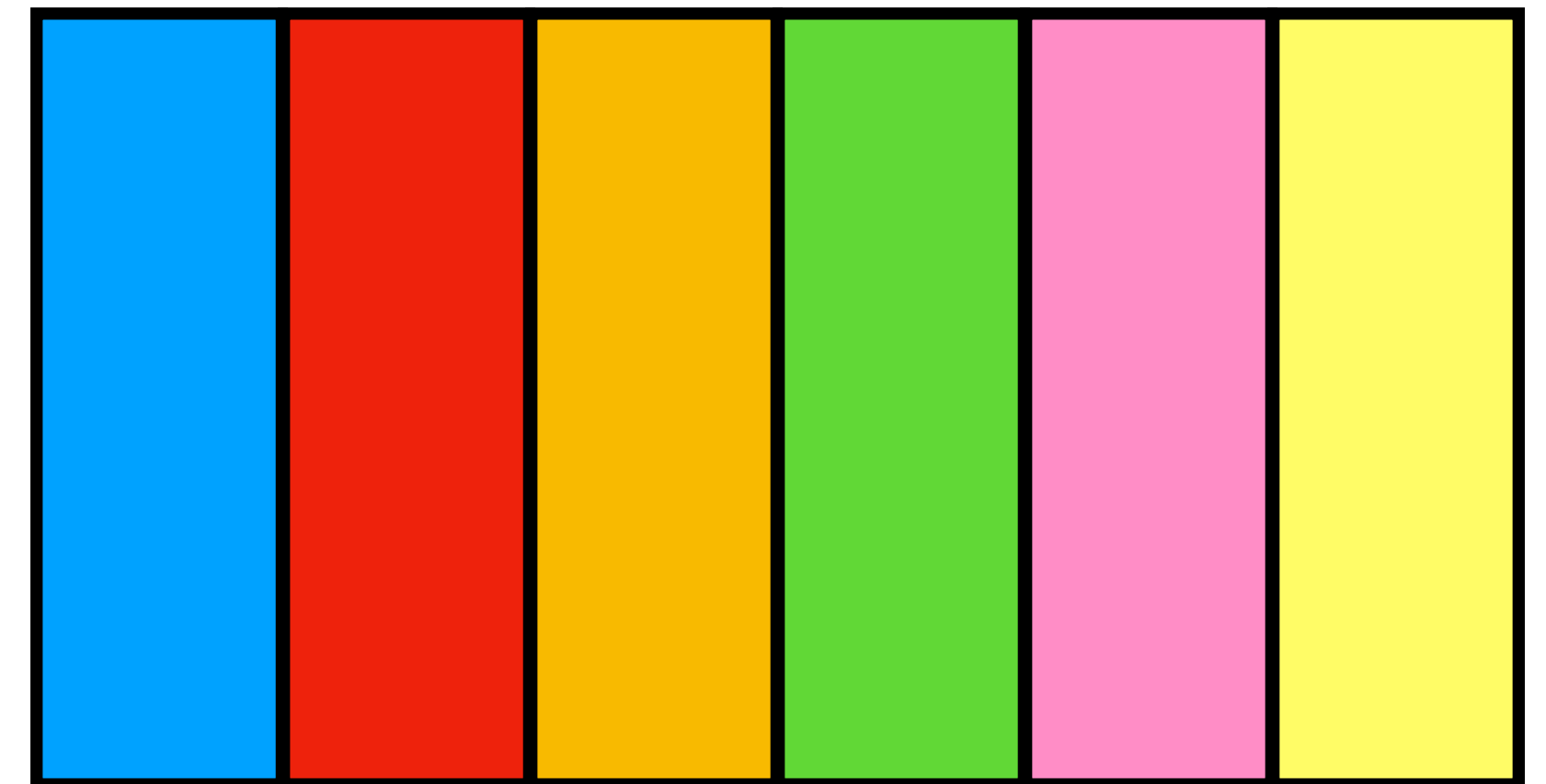
- Idea: add learnable position encoding vectors to input embedding vectors
- Alternative: a fixed sinusoidal encoding

Positional encoding



+

Sequence of vectors



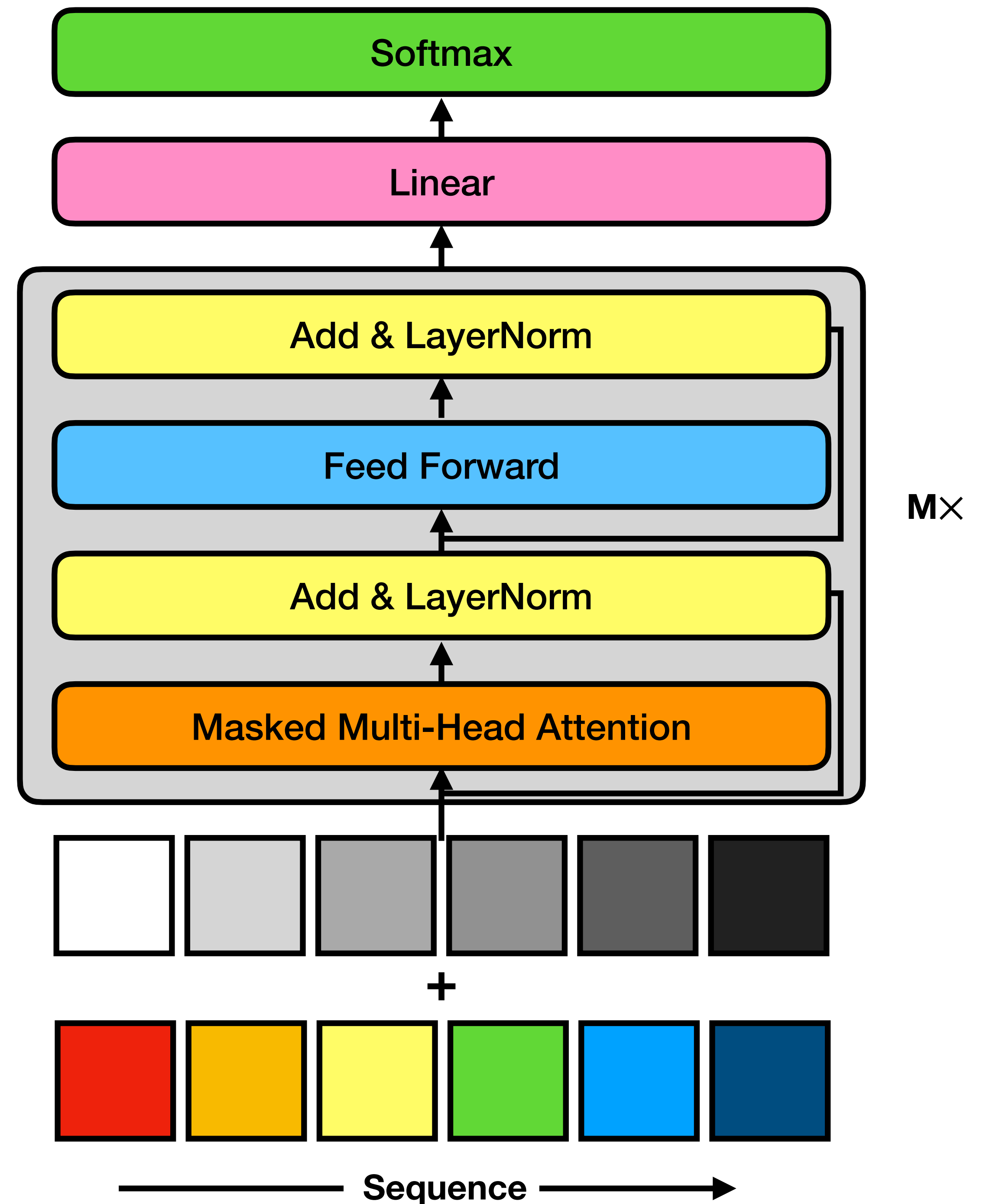
Decoder-only architecture

Output probabilities

Attention blocks

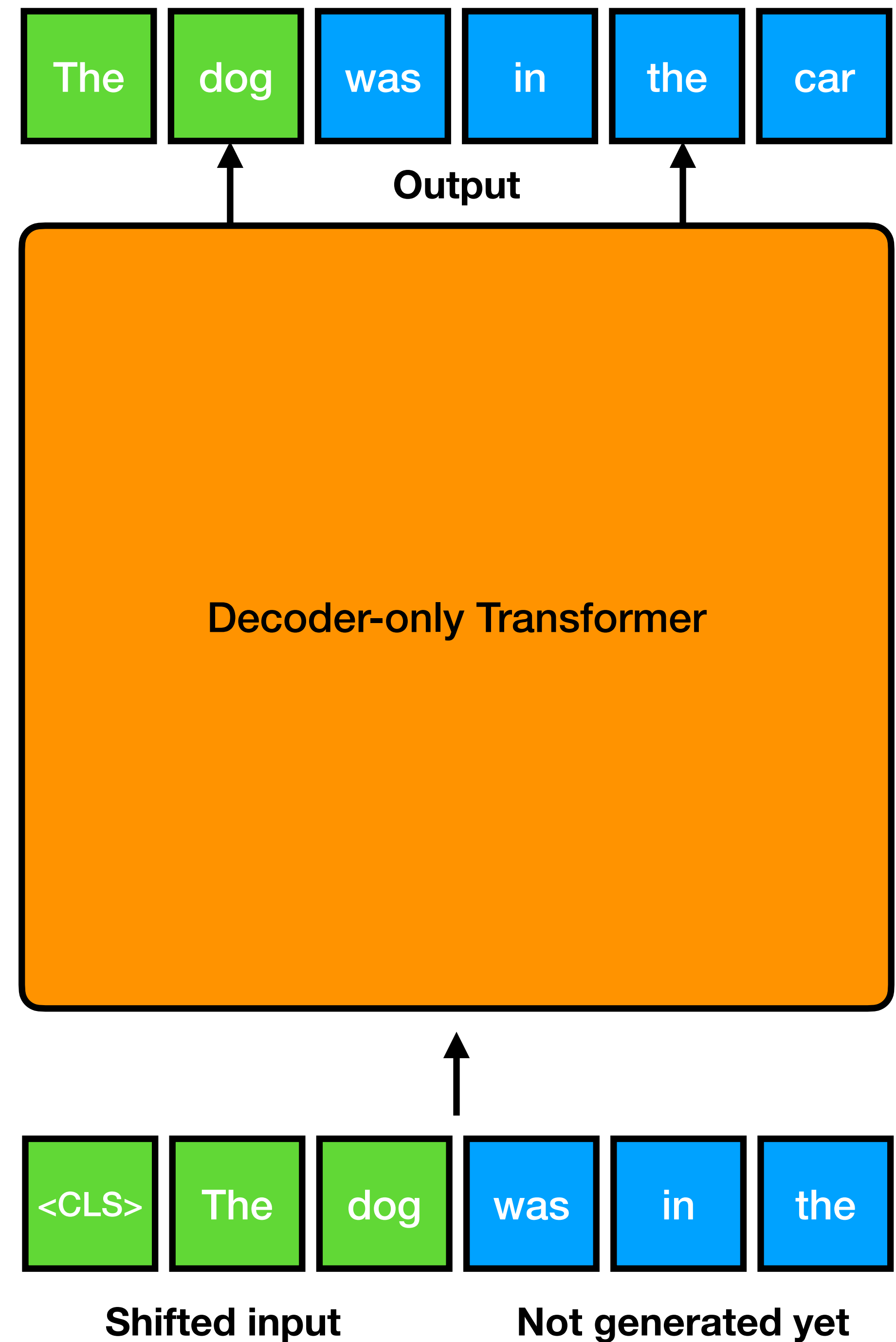
Positional encoding

Input tokens



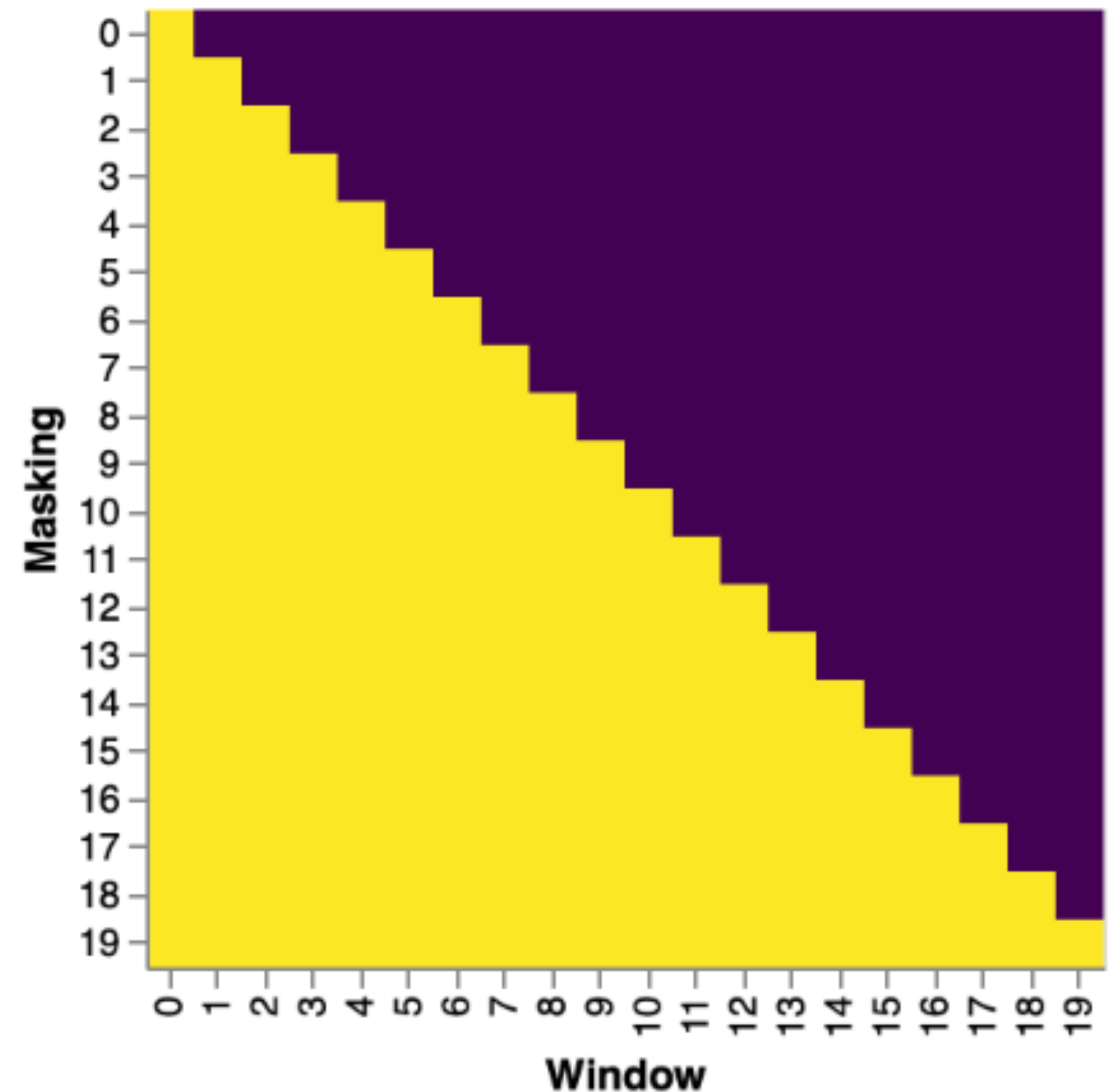
Sequence generation

Shifted input tokens + position encoding

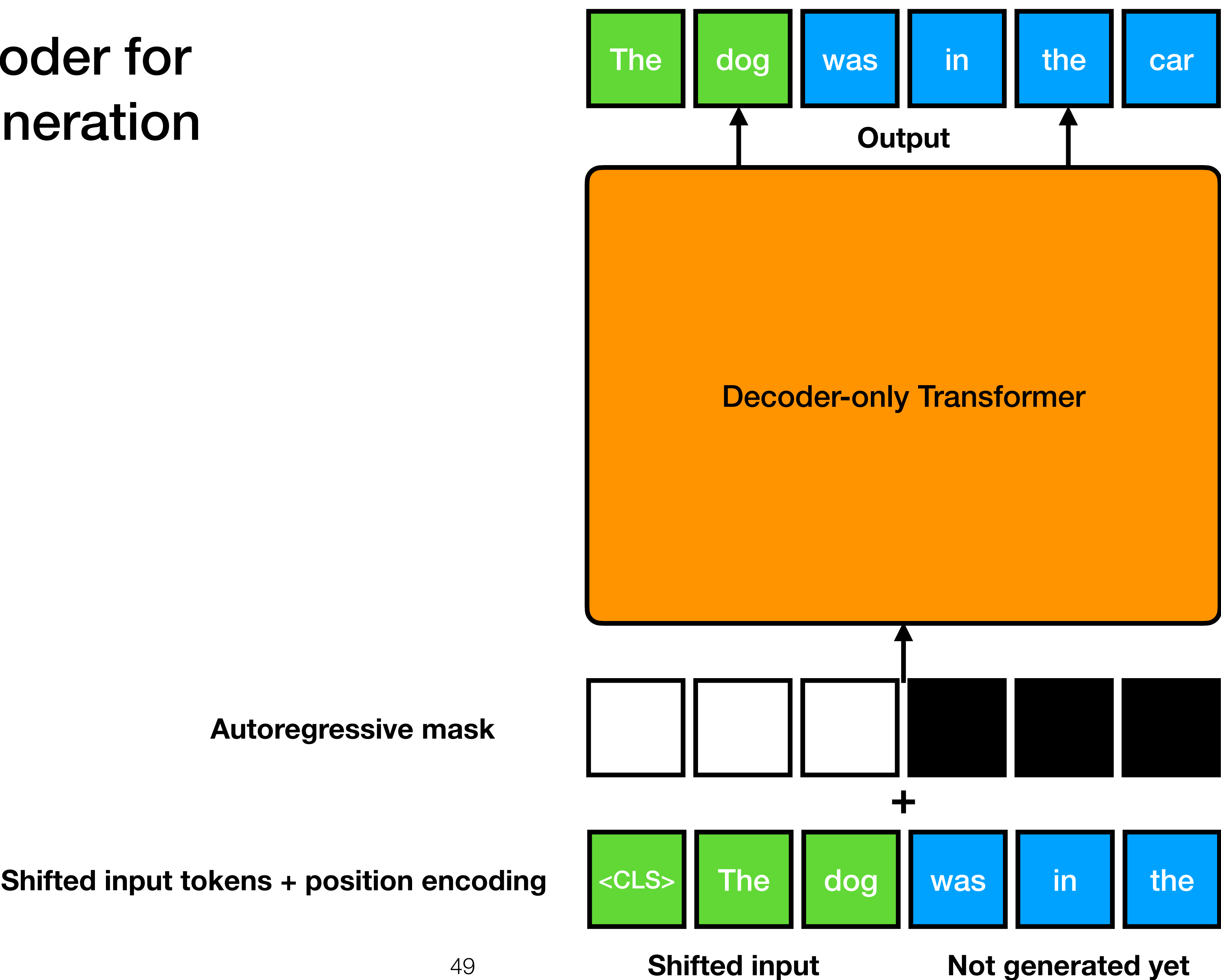


Masking

- When training a generative model, the attention block can “cheat” by looking ahead in the sequence. Why?
- This can be prevented by “masking” the input sequence.



Masked decoder for sequence generation



Computational Complexity

- Sequence length is N
- Size of embedding is d
- What is the cost of computing the following on the sequence (# multiplications):
 - MLP of width d
 - 1-D convolution of size k
 - Recurrent neural network
 - Self-attention